

High Order Discontinuous Galerkin Spectral Element Methods for Transitional and Turbulent Flow Simulations

Andrea D. Beck*, Thomas Bolemann, David Flad, Hannes Frank, Gregor J. Gassner,
Florian Hindenlang, Claus-Dieter Munz

*Institute for Aerodynamics and Gasdynamics
University of Stuttgart, Pfaffenwaldring 21, 70550 Stuttgart, Germany*

SUMMARY

In this paper, we investigate the accuracy and efficiency of Discontinuous Galerkin Spectral Method (DGSEM) simulations of under-resolved transitional and turbulent flows at moderate Reynolds numbers, where the accurate prediction of closely coupled laminar regions, transition and developed turbulence presents a great challenge to Large Eddy Simulation (LES) modelling. We take full advantage of the low numerical errors and associated superior scale resolving capabilities of high order spectral methods by using high order ansatz functions up to 12th order. We employ polynomial de-aliasing techniques to prevent instabilities arising from inexact quadrature of non-linearities. Without the need for any additional filtering, explicit or implicit modelling or artificial dissipation, our high order schemes capture the turbulent flow at the considered Reynolds number range very well.

Three classical LES benchmark problems are considered: A circular cylinder flow at $Re_D = 3900$, a confined periodic hill flow at $Re_h = 2800$ and the transitional flow over a SD7003 airfoil at $Re_c = 60,000$. For all computations, the total number of degrees of freedom used for the DGSEM simulations is chosen to be equal or considerably less than the reported data in literature. In all three cases, we achieve an equal or better match to Direct Numerical Simulation (DNS) results, compared to other schemes of lower order with explicitly or implicitly added subgrid scale models.

Copyright © 2013 John Wiley & Sons, Ltd.

Received . . .

KEY WORDS: Discontinuous Galerkin, under-resolved turbulence, transitional and detached flows, large eddy simulation, over-integration, stability, aliasing, polynomial dealiasing

1. INTRODUCTION

While variants of the Discontinuous Galerkin (DG) method were introduced by Reed and Hill as early as 1973 [65] and Lesaint and Raviart in 1974 [50] for the solution of linear transport equations, practical application of the method lay dormant for about two decades, until Cockburn

*Correspondence to: Institute for Aerodynamics and Gasdynamics
University of Stuttgart, Pfaffenwaldring 21, 70550 Stuttgart, Germany.
E-mail: beck/bolemann/flad/frank/gassner/hindenlang/munz@iag.uni-stuttgart.de

and Shu expanded the original ideas in a series of papers [25, 24, 23, 22] to systems of non-linear hyperbolic evolution equations. Bassi and Rebay developed discretization options for the diffusive terms [6] and extended the DG method to the compressible Navier-Stokes equations [8]. They also remarked on the need for consistent boundary discretization [7]. These efforts were accompanied by a number of researchers focusing on alternative approximations for the viscous fluxes [3, 2, 39, 51, 64], shock capturing [20, 45, 4] and spatial and temporal adaptation options [40, 44]. Beyond applications in inviscid or laminar problems (e.g. [9, 56]), there is a strong interest in extending DG to time-averaged turbulent flows in the form of the Reynolds Averaged Navier Stokes equations (RANS). Due to the implicit time stepping requirement for stationary problems and the associated memory load, the polynomial degree in these RANS simulations is usually moderate, i.e. between $N = 2$ and $N = 5$. Another major issue arising in adapting well-established RANS models to high order methods is the oscillatory behavior of the modeling equations, which can lead to severe stability issues [16]. Despite these difficulties, a number of successful high Reynolds number DG-RANS simulations have been reported, e.g. [59, 29, 5].

In contrast to this, the research activities into Discontinuous Galerkin methods for Large Eddy (LES) or Direct Numerical Simulations (DNS) of turbulent flows are less widespread. This is somewhat surprising, as in particular for well-resolved DNS scenarios, the low dispersion and dissipation properties of high order DG methods and its spectral character make it an attractive candidate. Also, the nature of these highly unsteady problems supports an explicit time integration approach, which in turn allows choosing very high order ansatz polynomials with associated low approximation errors. While continuous finite element spectral methods have been applied to incompressible DNS and LES problems with great success (e.g. [53]), the number of reported successful DG simulations for both incompressible and compressible cases is small. Collis [27] in 2002 was among the first to use high order DG methods for the DNS of compressible flows, and he applied a Smagorinsky eddy viscosity model to the small scales via the Variational Multiscale (VMS) method in a LES simulation [28]. More recently, Uranga et al. [70] used an $N = 3$ DG formulation for the simulation of a transitional flow over an airfoil, taking advantage of the numerical dissipation to account for the damping action of the unresolved scales in an implicit LES approach. A similar investigation was carried out by Carton de Wiart and Hillewaert [18]. The accuracy of Discontinuous Galerkin methods for high Reynolds number vortical flows for $N = 3$ was investigated in [19]. For higher orders ($N = 7 \dots 15$), Gassner and Beck [37] showed that stabilized Discontinuous Galerkin schemes applied to under-resolved simulations of isotropic turbulence compare very well with low order finite volume schemes with explicit or implicit subgrid scale models. In this case, the number of degrees of freedom (DOF) was chosen as in the finite volume approach, and no additional dissipation was added explicitly or implicitly to the DG formulation. Instead, the high resolution capabilities of high order discretizations resolve a much larger bandwidth of turbulent scales, thereby allowing the viscous action of the Navier-Stokes operator to remove most of the energy without the need for an additional model. In terms of computational efficiency, it was shown in [71] that high order DG schemes can compete with finite difference and finite volume formulations in a LES setting of this flow problem.

While these results in terms of accuracy and efficiency cannot be generalized to other problems, they indicate the potential of high order Discontinuous Galerkin discretizations as a baseline scheme for implicit or explicit Large Eddy simulations of vortex dominated flows. Especially for

LES of medium Reynold numbers flows as occur in biological systems or small unmanned aerial vehicles (UAV), where the energy content is localized in a relatively small wave number band, a high order discretization with associated high resolution per degree of freedom can be attractive, even without the need for an additional subgrid scale model.

In this paper, we continue and extend the work from [37] by investigating the applicability of high order DG schemes for transitional and turbulent flows. We therefore apply our method to well-established standard LES problems and assess our results in terms of accuracy and efficiency. A key ingredient in our approach is the de-aliasing mechanism, which avoids or reduces the approximation error associated with the numerical integration of the non-linear fluxes and thereby supports stable solutions on coarse discretizations. A fully de-aliased DG operator with sufficient integration of all inner products becomes independent of the choice of basis, and the only remaining parameters apart from the choice of the time integration method are the inviscid and viscous flux functions.

Recently, Malm et. al. [55] revisited the concept of polynomial de-aliasing in the context of a continuous spectral element method for the incompressible Navier-Stokes equations based on Legendre Gauss Lobatto nodes. They investigated the effect of additional filtering and a skew-symmetric operator to enforce the symmetry properties of the operator. In [35] and [36], skew symmetric formulations for nodal Discontinuous Galerkin Spectral Element approximations are introduced as well.

In our approach, no additional filtering or added dissipation is included. While from a discretization standpoint, a very high order ansatz is highly attractive, two issues have to be taken into consideration: (i) the generation of the corresponding high order meshes is far from trivial and an ongoing subject of research [41], (ii) a high order discretization implies (for a fixed number of degrees of freedom) a very coarse grid. Without sub-element parallelization, the number of elements thus constitutes an upper bound on the number of computing nodes and therefore on the total simulation time.

1.1. Goal of this paper

In this paper, we demonstrate that high order DG schemes are very well suited for the simulation of transitional and turbulent flows at moderate Reynolds numbers. We employ the same number of degrees of freedom (or less) as published for established LES schemes, and we show that we can achieve the same or better agreement with DNS data. In these simulations, we do not require any implicit or explicit LES modelling term, but use a standard DG scheme. We will also show that due to the efficient parallelization of this approach, we can run the full simulations in a matter of hours.

1.2. Structure of this paper

The structural organization of this paper is as follows: In Sec. 2, we describe the Discontinuous Galerkin Spectral Element Method and give details on the de-aliasing strategy in Sec. 2.2 and Sec. 2.3. In Sec. 3, we describe our simulation of the subcritical flow over a circular cylinder at $Re_D = 3900$, where transition to turbulence occurs in the laminar shear layer. In Sec. 4, we report the setup and results of the periodic hill flow. The flow over an airfoil at $Re_c = 60,000$ in Sec. 5

is the most demanding of these cases, as it includes laminar separation, shear layer transition and turbulent reattachment. We end with a summary and a conclusion in Sec. 6.

2. NUMERICAL METHODS

In this work we consider the compressible Navier-Stokes equations (NSE) expressed in conservation form

$$U_t + \vec{\nabla}_x \cdot \vec{F}(U, \vec{\nabla}_x U) = 0, \quad (1)$$

where U denotes the vector of conserved quantities $U = (\rho, \rho v_1, \rho v_2, \rho v_3, \rho e)^T$, the subscript t the time derivative and $\vec{\nabla}_x$ the gradient operator in physical space. The flux is the difference of advection and viscous fluxes, $\vec{F} = \vec{G}(U) - \vec{H}(U, \vec{\nabla}_x U)$, and consists of the columns

$$G^i(U) = \begin{pmatrix} \rho v_i \\ \rho v_1 v_i + \delta_{1i} p \\ \rho v_2 v_i + \delta_{2i} p \\ \rho v_3 v_i + \delta_{3i} p \\ \rho e v_i + p v_i \end{pmatrix}, \quad H^i(U, \vec{\nabla}_x U) = \begin{pmatrix} 0 \\ \tau_{1i} \\ \tau_{2i} \\ \tau_{3i} \\ \tau_{ij} v_j - q_i \end{pmatrix} \quad (2)$$

with $i = 1, 2, 3$. We follow the usual nomenclature for $\rho, (v_1, v_2, v_3)^T, p, e$ denoting the density, velocity vector, pressure and specific total energy. The viscous stress tensor $\underline{\tau}$ appearing in $\vec{H}(U, \vec{\nabla}_x U)$ is defined as

$$\underline{\tau} := \mu(\vec{\nabla}_x \vec{v} + (\vec{\nabla}_x \vec{v})^T) - \frac{2}{3}(\vec{\nabla}_x \cdot \vec{v})\underline{I}, \quad (3)$$

and the heat flux as $\vec{q} := (q_1, q_2, q_3)^T$ with

$$\vec{q} = -k\vec{\nabla}_x T, \quad (4)$$

where $k = \frac{c_p \mu}{Pr}$. Fluid dependent variables are heat conductivity, viscosity coefficient, Prandtl number and the specific heats denoted by k, μ, Pr, c_p and c_v . These variables are assumed to be constant. Using the adiabatic coefficient $\kappa = \frac{c_p}{c_v}$ and the specific gas constant $R = c_p - c_v$, the perfect gas law

$$p = \rho R T = (\kappa - 1)\rho(e - \frac{1}{2}\vec{v} \cdot \vec{v}), \quad e = \frac{1}{2}\vec{v} \cdot \vec{v} + c_v T \quad (5)$$

is used to close the system of equations.

2.1. Discontinuous Galerkin Spectral Element Method

In order to solve the system of equations, the computational domain is subdivided into non-overlapping hexahedral elements, which we allow to be connected in a fully unstructured, but conforming way. In a first step, the NSE are transformed to a reference element $E \in [-1, 1]^3$. The associated mapping function $\vec{x}(\vec{\xi})$ is used to calculate the Jacobian determinant $J(\vec{\xi}) = \det(\frac{\partial \vec{x}}{\partial \vec{\xi}})$, for brevity we drop the dependence on $\vec{\xi}$ in the following. Kopriva [46] showed how to construct these metrics ensuring the so called free-stream preserving property. Applying the transformation

to Eq. (1) leads to

$$JU_t + \vec{\nabla}_\xi \cdot \vec{\mathcal{F}}(U, \vec{\nabla}_x U) = J(\vec{\xi}) U_t + \vec{\nabla}_\xi \cdot (\vec{\mathcal{G}}(U) - \vec{\mathcal{H}}(U, \vec{\nabla}_x U)) = 0, \quad (6)$$

with the transformed fluxes $\vec{\mathcal{F}}$. For our scheme we employ a tensor product of 1-D Lagrange polynomials ℓ^N of degree N to express the discrete approximate solution vector as

$$U(\vec{\xi}, t) = \sum_{i,j,k=0}^N \hat{U}_{ijk}(t) \psi_{ijk}^N(\vec{\xi}), \quad \psi_{ijk}^N(\vec{\xi}) = \ell_i^N(\xi^1) \ell_j^N(\xi^2) \ell_k^N(\xi^3), \quad (7)$$

where $\hat{U}_{ijk}(t)$ are time dependent nodal degrees of freedom. Following Kopriva [47], the $N + 1$ Gauss-Legendre quadrature points $\{\xi_i\}_{i=0}^N$ are chosen as interpolation nodes. Analogously, the discrete transformed flux $\vec{\mathcal{F}}$ reads as

$$\mathcal{F}^l(\vec{\xi}) \approx \sum_{i,j,k=0}^M \hat{\mathcal{F}}_{ijk}^l \psi_{ijk}^M(\vec{\xi}), \quad l = 1, 2, 3 \quad (8)$$

$$\hat{\mathcal{F}}_{ijk}^l = \mathcal{G}(U) - \mathcal{H}(U, \vec{\nabla}_x U) \big|_{\vec{\xi}_{ijk}} \quad (9)$$

with $\psi_{ijk}^M(\vec{\xi}) = \ell_i^M(\xi^1) \ell_j^M(\xi^2) \ell_k^M(\xi^3)$. Note that the nodal fluxes are evaluated on $M + 1$ Gauss-Legendre quadrature points, with $M \geq N$. This implementation allows for polynomial de-aliasing of the non linear flux terms [43]. The choice of M depends on the non-linearity of the flux for under-resolved calculations, more information on this topic is given in Sec. 2.2.

Multiplying Eq. (6) by the test function $\phi(\vec{\xi})$ and integrating over the reference element leads to

$$\int_E \left(JU_t + \vec{\nabla}_\xi \cdot \vec{\mathcal{F}}(U, \vec{\nabla}_x U) \right) \phi(\vec{\xi}) d\vec{\xi} = 0. \quad (10)$$

Following the Galerkin approach the test functions are chosen identical to the basis functions in Eq. (7). Using a spatial integration by parts yields the weak formulation

$$\frac{\partial}{\partial t} \int_E JU \phi d\vec{\xi} + \oint_{\partial E} (\mathcal{G}_n^* - \mathcal{H}_n^*) \phi ds - \int_E \vec{\mathcal{F}}(U, \vec{\nabla}_x U) \cdot \vec{\nabla}_\xi \phi d\vec{\xi} = 0, \quad (11)$$

where $\mathcal{G}_n^*$ denotes the surface normal numerical flux function for the inviscid terms, given by $\mathcal{G}_n^* := \mathcal{F}_n^*(U^+, U^-)$ and superscripts $\pm$ denote the values at the grid cell interface from the neighbor and the local grid cell, respectively. For the inviscid numerical flux, several well-known flux functions are implemented, which ensure consistency and uniqueness of the numerical flux. In this work we apply Lax-Friedrichs as well as Roe's approximate Riemann solver flux approximations. For details on those Riemann solvers we refer to the textbook by Toro [69]. $\mathcal{H}_n^*$ denotes the numerical flux function for the viscous term, resulting from the viscous flux $\vec{\mathcal{H}}$ in Eq. (6).

Through this viscous flux, Eq. (11) depends on the solution gradient $\vec{\nabla}_x U$. The treatment of these terms in the context of DG approximations was first tackled by Bassi and Rebay [10, 6], who introduced a mixed finite element approximation. The governing equations Eq. (1) are rewritten as

a corresponding system of first order equations with auxiliary variable $\vec{S}$

$$\begin{aligned}\vec{S} - \vec{\nabla}_x U &= 0, \\ U_t + \vec{\nabla}_x \cdot \vec{F}(U, \vec{S}) &= 0.\end{aligned}\tag{12}$$

Applying the discretization steps outlined above to the auxiliary equation leads to

$$\begin{aligned}c = 1, \dots, 5 : \quad & \int_E J \vec{s}_c \phi d\vec{\xi} + \oint_{\partial E} \vec{u}_{c,n}^* \phi ds - \int_E u_c \cdot \vec{\nabla}_\xi \phi d\vec{\xi} = 0, \\ & \frac{\partial}{\partial t} \int_E JU \phi d\vec{\xi} + \oint_{\partial E} (\mathcal{G}_n^* - \mathcal{H}_n^*) \phi ds - \int_E \vec{F}(U, \vec{S}) \cdot \vec{\nabla}_\xi \phi d\vec{\xi} = 0,\end{aligned}\tag{13}$$

with the component u_c of the state vector U and its lifting operator $\vec{s}_c$. The numerical flux of the auxiliary equation is $\vec{u}_{c,n}^*$, and $\mathcal{H}_n^* = \mathcal{H}_n^*(U^+, U^-, \vec{S}^+, \vec{S}^-)$ denotes the numerical flux function for the viscous terms. Following [6], we choose

$$c = 1, \dots, 5 : \quad \vec{u}_{c,n}^* = \frac{1}{2} (u_c^+ + u_c^-) \vec{n},\tag{14}$$

$$\mathcal{H}_n^* = \frac{1}{2} \left(\mathcal{H}_n(U^+, \vec{S}^+) + \mathcal{H}_n(U^-, \vec{S}^-) \right),\tag{15}$$

with $\vec{n}$ denoting the outward pointing surface normal. This treatment of the viscous fluxes is usually labeled BR1 and is known to be formally unstable for purely elliptic problems. However, our simulations focus on advection dominated problems, hence the inviscid numerical flux function is helping in stabilizing the overall operator and no issues have been observed in any of our computations.

Due to the weak coupling of the elements through fluxes, it is an obvious idea to enforce the boundary conditions in the same manner, even for the Dirichlet type. The rationale for this approach is to ensure consistency in the approximation of the solution and the boundary conditions. Collis [27] investigated the effect of weakly versus strongly imposed Dirichlet conditions for the case of an underresolved one-dimensional stationary boundary layer problem and turbulent channel flows. He showed that the solution quality measured in the L_2 and H_1 error norms is greatly enhanced when using a weak application of the boundary conditions, while the L_∞ norm slightly favors the strong form, as this norm reacts directly to the error directly at the boundary.

We follow this approach, and we enforce the boundary conditions for Eq. (13) weakly through the prescription of the boundary fluxes $\vec{u}_{c,n}^*$, $\mathcal{G}_n^*$ and $\mathcal{H}_n^*$. While Collis reported improvements using weakly enforced wall boundary conditions, we also have noted positive effects at freestream boundaries: reflections stemming from outgoing acoustic waves are hardly encountered. Even in aeroacoustic computations, where one main simulation goal is an unpolluted instantaneous pressure field, this treatment was found to be sufficient.

After numerically integrating Eq. (13) by Gauss quadrature in space, we obtain a semi discrete form, which then is advanced in time by the fourth order accurate explicit Runge-Kutta method of Carpenter and Kennedy [17].

2.2. Stable Simulation of Under-resolved Flows

In section 2.1 we introduced the transformed flux function in Eq. (8) as

$$\mathcal{F}^l(\vec{\xi}) \approx \sum_{i,j,k=0}^M \hat{\mathcal{F}}_{ijk}^l \psi_{ijk}^M(\vec{\xi}), \quad l = 1, 2, 3 \quad (16)$$

$$\hat{\mathcal{F}}_{ijk}^l = \mathcal{G}(U) - \mathcal{H}(U, \vec{\nabla}_x U) \big|_{\vec{\xi}_{ijk}} \quad (17)$$

We note that the non-linearity of the advection part of the flux $\mathcal{F}^l(U)$ is either quadratic for incompressible flows or cubic in a compressible setting. Thus, a collocation of the flux $\mathcal{F}^l$ on $(N+1)^3$ solution points $\vec{\xi}_{ijk}$, which in our implementation double as integration nodes, introduces an error in the interpolant (Eq. (8)) and subsequently in the associated integration, the volume contribution of Eq. (11):

$$\begin{aligned} \int_E \mathcal{F}^l(U, \nabla_x U) \frac{\partial}{\partial \xi^l} \phi(\vec{\xi}) d\vec{\xi} &\approx \sum_{p,q,r=0}^M \sum_{i,j,k=0}^M \hat{\mathcal{F}}_{ijk}^l \psi_{ijk}^M(\vec{\xi}_{pqr}^M) \frac{\partial}{\partial \xi^l} \phi(\vec{\xi}) \bigg|_{\vec{\xi}=\vec{\xi}_{pqr}^M} \omega_p^M \omega_q^M \omega_r^M \\ &= \sum_{p,q,r=0}^M \hat{\mathcal{F}}_{pqr}^l \frac{\partial}{\partial \xi^l} \phi(\vec{\xi}) \bigg|_{\vec{\xi}=\vec{\xi}_{pqr}^M} \omega_p^M \omega_q^M \omega_r^M \end{aligned} \quad (18)$$

where we have introduced the approximation of the integral through a tensor product of one-dimensional Gauss quadratures with $(M+1)^3$ associated nodes $(\vec{\xi}_{pqr}^M)$ and weights $(\omega_p^M, \omega_q^M, \omega_r^M)$. Since $\mathcal{F}^l$ is at most a polynomial of degree $3N$ for compressible flows, there exist values of M for which the quadrature of the right hand side of Eq. (18) becomes exact. From the exactness of the quadrature rule, M is found to be $2N$ for cubic and $\frac{3}{2}N$ for quadratic non-linear integrands. Consequently, an inexact integration of these terms, i.e. an insufficient choice of M , results in an approximation error often termed “aliasing”. For under-resolved simulations where the modal content in the higher modes is not negligible, these aliasing errors accumulate energy in the high modes, and eventually lead to an instability of the scheme. Based on the general idea of de-aliasing by exact projection developed by Orszag [61], Kirby and Karniadakis showed in [43] that de-aliasing for polynomial spectral methods (often called over-integration in this context) is achieved similarly by choosing a sufficient integration precision with respect to the ansatz function. It should be noted that for lower order DGSEM, under-resolved simulations may remain stable in many cases even without de-aliasing, since these lower order schemes are dominated by dissipation over a broad range of scales, which may cover any aliasing errors.

Eq. (11) highlights another important aspect of the de-aliasing operation: once the inner products are evaluated exactly through sufficient quadrature, the only remaining parameters in the spatial discretization are the viscous and inviscid numerical flux choices $\mathcal{H}_n^*$ and $\mathcal{G}_n^*$. Thus, through exact integration, the “original” DG scheme and its properties can be recovered and analyzed, without interference from the choices of discretization. Therefore, this formulation is particularly attractive for the evaluation of the DG scheme, the effects of LES subgrid scale models or the choice of the flux functions [11].

Since the use of over-integration prevents numerical instabilities through wrongfully represented (i.e. aliased) energy, the only issue remaining is the lack of sub grid scales and their associated

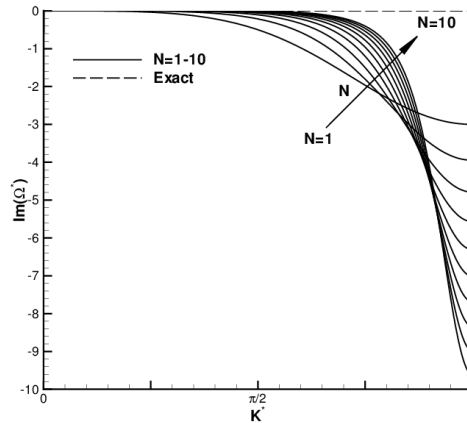


Figure 1. Dissipation relation of DGSEM with Gauss nodes, from $N = 1$ to $N = 10$. $K^* = \frac{K}{N+1}$ is the normalized wavenumber and Ω^* is the corresponding modified normalized wavenumber. Reproduced from [38] with permission.

dissipation, as a direct consequence of under-resolution. Thus a mechanism needs to be present that accounts for the energy transferred to high wavenumbers, or in other words, mimics the dissipation of the unresolved scales, while leaving large scales unaffected. In a classical LES setting, this role is played by the (explicitly added or implicitly introduced) subgrid modelling terms. In our approach, we do not add any type of model or filter, instead we rely on the combination of two properties of the high order schemes: the wide bandwidth of resolved scales, which carry a non-negligible physical dissipation, and the dissipation properties of the scheme. Gassner and Kopriva [38] showed that with increasing polynomial degree the onset of dissipation errors within a reference element due to insufficient resolution is pushed towards higher wavenumbers (Fig. 1). Using Gauss points, a spectral cut-off like dissipation character of the DG operator is obtained with increasing polynomial degree, i.e. small errors for a wide range of resolved scales and high dissipation for the marginally represented wavenumbers, a property which constitutes the robustness of the scheme. The potential of high order schemes for under-resolved simulation has been studied for the Taylor-Green-Vortex flow by Gassner and Beck in [37], showing that the solution accuracy is strongly enhanced by applying high order methods while leaving the total number of degrees of freedom (DOF) constant. It was pointed out that this effect is caused by the discussed low dissipation and dispersion error for a wide range of resolved scales.

2.3. Numerical Strategy

In summary, our approach for under-resolved turbulence simulations is characterized as follows. Driven by the need for an accurate resolution over a wide range of scales unpolluted by numerical errors, we use a method with a high polynomial degree. Practical limits are mesh generation and parallelization issues. In order to prevent integration crimes, we apply over-integration with as many integration points as necessary to achieve a stable simulation. While the theoretical integration requirements for quadratic and cubic non-linear functions were discussed above, we find that in practical cases, these requirements can be relaxed. We note that our scheme does not use

Table I. Integral quantities and simulation parameters for $Re_D = 3900$ cylinder flow

<i>Author</i>	C_{pBase}	Str	C_D	L_r/D	<i>Scheme</i>	<i>LES</i>	<i>DOF</i>
Kravchenko et al. [48]	-0.94	0.210	1.04	1.35	B-Spline SEM	Smag.	1-2 M
Blackburn et al. [13]	-0.93	0.218	1.01	1.63	GL-SEM	Smag.	1.5 M
Meyer et al. [58]	-0.92	0.210	1.05	1.38	FV	ALDM	6.0 M
Fröhlich et al. [33]	-1.03	0.216	1.08	1.09	FV	Smag.	1.4 M
Ouvrard et al. [62]	-0.85	0.218	0.99	1.54	FV/FE	Smag.	1.5 M
Ouvrard et al. [62]	-0.81	0.226	0.93	1.68	FV/FE-VMS	Smag	1.5 M
Franke et al. [31]	-0.85	0.209	0.98	1.64	FV	Yoshizawa	1.2 M
Ma et al. (I) [54]	-0.96	0.203	0.96	1.12	h/p-FEM	DNS	24 M
Ma et al. (II) [54]	-0.84	0.219	0.96	1.59	Fourier spanw. h/p-FEM Fourier spanw.	DNS	12 M
Current: $N = 11$	-1.01	0.206	1.10	1.13	DGSEM	-	0.7 M
Current: $N = 7$	-0.91	0.208	1.00	1.31	DGSEM	-	1.6 M

C_{pBase} : pressure coefficient at the downstream position $x = D/2, y = 0$, Str : Strouhal number of the lift coefficient $f_{Lift} D/u_\infty$, C_D : drag coefficient, L_r : length of separation bubble

any filtering, additional artificial viscosity or subgrid scale modelling terms, which all introduce additional sources of error.

3. SUBCRITICAL FLOW OVER A CIRCULAR CYLINDER

3.1. Computational Setup

The flow over a cylinder with circular cross section at $Re_D = 3900$ is a well-established test case for the evaluation of implicit and explicit LES modelling approaches and has been extensively investigated by a large number of researchers [48, 13, 58, 33, 62, 31, 54]. The Reynolds number based on the cylinder diameter D is defined as $Re_D = (u_\infty \rho_\infty D)/\mu$, with the freestream velocity U_∞ , the freestream density ρ_∞ and the kinematic viscosity $\nu = \mu/\rho$. At the given flow conditions, the boundary layer remains laminar until the separation point, where the shear layer is shed periodically. The ensuing transition to turbulence occurs very close to the geometry in the shear layer, and interacts with the resulting reverse flow region. Due to this complex flow situation, this test case proves challenging for the LES methods, which explains the considerable spread in reported results. Tab. I collects the integral quantities of LES and DNS simulations of this flow from literature.

3.1.1. Spatial Discretization While our code supports fully unstructured hexahedral meshes, we choose to discretize this simple geometry in a structured manner with a fully volume-curved mesh, i.e. the mapping $\vec{x}(\vec{\xi})$ is a polynomial of arbitrary degree, which can be chosen to match the polynomial ansatz for the unknown flow variables to ensure a consistent geometry discretization (called an isoparametric mapping). The flow domain in a plane perpendicular to the cylinder axis ($x - y$ plane) is circular, with the origin located at the center of the geometry and a radial extension of $r = 100D$. In a recent comparative study, Lysenko et al [52] report the radial domain extension of 12 LES computations of this cylinder flow. They vary between $r = 10D$ for the smallest and

$r = 35D$ for the largest domains. For radial domain sizes between $r = 10D$ and $r = 50D$, we found an influence on the pressure field through a choking effect when using Dirichlet free stream boundary conditions.

Regarding the spanwise extension Δz of the flow domain, it should be noted that Ma et al. [54] reported a strong dependence of the integral results and the shape of the velocity deficit in the wake. The spanwise domain sizes in the literature results presented in Tab. I vary from πD to $2\pi D$. The two DNS results according to Ma et al. correspond to $\Delta z = 2\pi D$ and $\Delta z = \pi D$. In our computations, we choose a spanwise extension Δz of $4D$, following [58], which is discretized by an equidistant mesh. In the axial and tangential direction ($x - y$ plane), we choose an exponential stretching in the radial and a Gaussian stretching in the circumferential direction to cluster the grid points towards the geometry and in the wake. Using again the reported data in Tab. I as a reference, we limit the number of degrees of freedom of our computations to a maximum of about 1.5 million. Following our strategy outlined in Sec. 1, we choose two high order discretization approaches: firstly, an $N = 7$ setup, where N denotes the degree of the one-dimensional polynomial ansatz basis, leading to $(N + 1)^3 = 512$ degrees of freedom per element and a total of 1.57 M DOF on the associated grid. For an exact integration, we choose $M = 11$ in accordance with the integration rule for quadratic non-linear functions, as we set the Mach number Ma for an essentially incompressible flow (see Sec. 3.1.2). The second discretization consists of an ansatz of $N = 11$ polynomials with $M = 17$, giving a total of only 0.66 M. on the associated grid. Tab. II collects the details of the two discretizations used. A zoomed-in view of both grids and the resulting time and spanwise averaged streamwise velocity distribution is shown in Fig. 2. We use the local Lax-Friedrichs flux for the convective fluxes, and Bassi and Rebays first method BR1 for the viscous contributions.

3.1.2. Boundary Conditions and Data Acquisition All computations are started from constant free-stream initial conditions in the whole domain. We choose the density as $\rho_\infty = 1.0$, the freestream velocity vector as $\vec{u}_\infty = (1, 0, 0)^T$ and the circular diameter of the cylinder to be $D = 1.0$. The required Reynolds number is set via the kinematic viscosity ν . We choose a Mach number $Ma = 0.1$ to compare with the incompressible data published in literature. This (together with an adiabatic exponent $\gamma = 1.4$ for air) leads to a stagnation temperature of $1.002 T_\infty$, which is used as the wall temperature for the isothermal cylinder wall boundary conditions. In the spanwise direction, periodic boundary conditions are enforced, while the remaining domain boundaries are set to weakly enforced Dirichlet free stream conditions. We did not encounter any reflection of acoustic waves at the downstream boundary, and no upstream propagating disturbances were visible in the dilatation or pressure fields.

Fig. 3 shows the temporal development of both the streamwise and cross flow force coefficient for the $N = 7$ computation. It is interesting to note that the de-aliased schemes (both $N = 7$ and $N = 11$) are stable through the initial unphysical transient from the constant state to the onset of vortex shedding, although very strong non-linear interactions go along with this adjustment. This contradicts the general notion that high order schemes lack robustness in practical simulations.

The collection of the time averaged data was started after the establishment of stable vortex shedding at $100 T^*$, with the convective time unit T^* defined as D/u_∞ and continued for 80 shedding cycles. In a postprocessing step, the gathered time averaged fields were averaged in the spanwise direction.

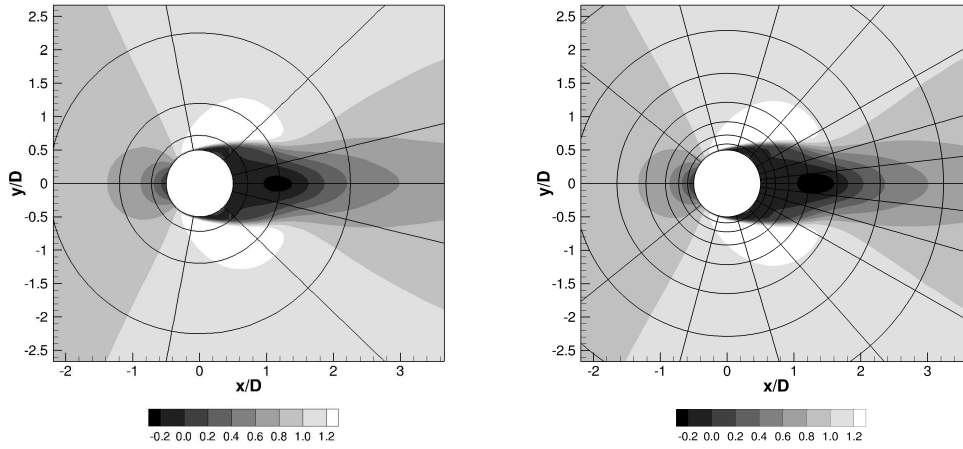


Figure 2. Time and spanwise averaged streamwise velocity component and grid. *left*: $N = 11$ grid with $8 \times 8 \times 6$ elements (circumferential, radial and spanwise), *right*: $N = 7$ grid with $16 \times 16 \times 12$ elements.

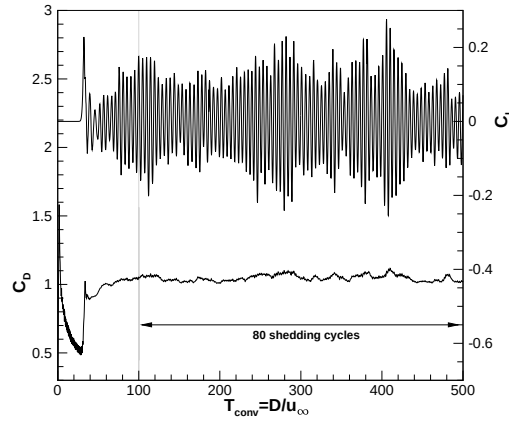


Figure 3. Temporal development of the lift and drag coefficients of the $N = 7$ computation.

Table II. Degrees of freedom and mesh details of the cylinder flow simulations.

Case	no. DOF	elements circum.	elements radial	elements spanwise	N	M
$N = 7$	1.573 M	16	16	12	7	11
$N = 11$	0.663 M	8	8	6	11	17

3.2. Results

Along with the data reported for LES with implicit and explicit modelling and DNS results, Tab. I lists the results for the integral flow quantities of our high order computations. We compare the base pressure $C_{p_{Base}}$, computed as $(p - p_\infty)/(0.5\rho_\infty u_\infty^2)$, the Strouhal number of the lift force from $\frac{f_{Lift} D}{u_\infty}$, the drag coefficient C_D (computed from the streamwise force and with the projected cylinder surface in that direction) and the normalized length of the recirculation bubble L_r/D . In general, our integral quantities agree well with the published data, while the match for $N = 11$ case is the weaker of the two. This can be attributed to the very low total number of degrees of freedom

for this case and the insufficient wall resolution. Still, even for this severely under-resolved situation, the results are in reasonable agreement.

The base pressure coefficient or pressure recovery coefficient $C_{p_{Base}}$ contributes strongly to the total drag of the cylinder. Fig. 4 *right* shows its distribution along the cylinder surface. We also report DNS data from Ma et al. [54] and experimental measurements for a slightly higher Reynolds number from Norberg, taken from the publication by Kravchenko and Moin [48]. We note a perfect agreement to the DNS results in the acceleration region on the upstream face of the cylinder, with a slight deviation in the separation region. The agreement in the reverse flow region at the downstream face of the cylinder is again very close to both experimental and DNS data, with the $N = 7$ results matching the measurements almost perfectly.

Fig. 4 *left* corroborates these observations. Here, we have shown the spanwise vorticity component on the cylinder surface Ω_z , normalized by $\sqrt{0.5 RE}$. We reproduce DNS results from Ma et al. [54] along with experimental data for $Re = 5000$, measured by Son & Hanratty (taken from [54]). The $N = 7$ and $N = 11$ results are in very good agreement with the DNS results, and show only a small deviation in the separation region. The sign reversal in our results indicates the existence of two separate separation bubbles with intermediate reattachment, a feature not found in the DNS. This behavior is also visible as the plateau region in the pressure coefficient (Fig. 4 *right*). On the downstream side of the cylinder, the match between the current results and the DNS is again very good.

In Fig. 5 and 6, we show the time and spanwise averaged streamwise and crossflow velocities $\langle u \rangle$ and $\langle v \rangle$, normalized by the freestream velocity, at different locations downstream of the geometry. We compare our profiles with experimentally determined values from Parnaudeau et al. [63], and two LES computations: The implicit approach through an approximate deconvolution method used in conjunction with a low order Finite Volume formulation presented in [58] and a high order B-Spline based Galerkin projection method with explicitly added Smagorinsky model [48]. The evolution of the wake profile is of particular interest, as the mixing and recovery of the velocity deficit is governed by viscous action, and therefore by the combination of resolved and numerical viscosity. A high viscosity generally leads to a fast interchange of momentum through shear and thus to a less pronounced but overall longer wake.

The agreement of the current results for the streamwise velocity $\langle u \rangle$ (Fig. 5) is very good for all locations in the wake for the $N = 7$ case. In particular, a very close agreement with the data of Kravchenko and Moin can be observed. Only at the farthest location $x/d = 2.02$, a slight upward shift of the current results is visible. This is most likely due to the added Smagorinsky viscosity in the Kravchenko and Moin computations, which leads to a slightly longer wake region. For the $N = 11$ case, the match with the published data is again very good in the near wake, but deteriorates downstream. Together with the data in Tbl. I which indicates a higher drag coefficient and reduced length of the separation bubble, this behavior is characteristic of too little dissipation. The notably reduced spatial resolution and the additional reduction in numerical dissipation from an $N = 7$ to an $N = 11$ approximation result in this compact, high drag wake. This motivates the addition of subgrid scale dissipation for higher Reynolds numbers or reduced spatial resolution.

For the cross flow velocity $\langle v \rangle$ (Fig. 6), we observe a similar behavior. The general match with both experimental and LES data for the $N = 7$ case is very good, although the agreement with the Kravchenko and Moin data [48] is weaker further downstream. The match with Meyer and Hickel

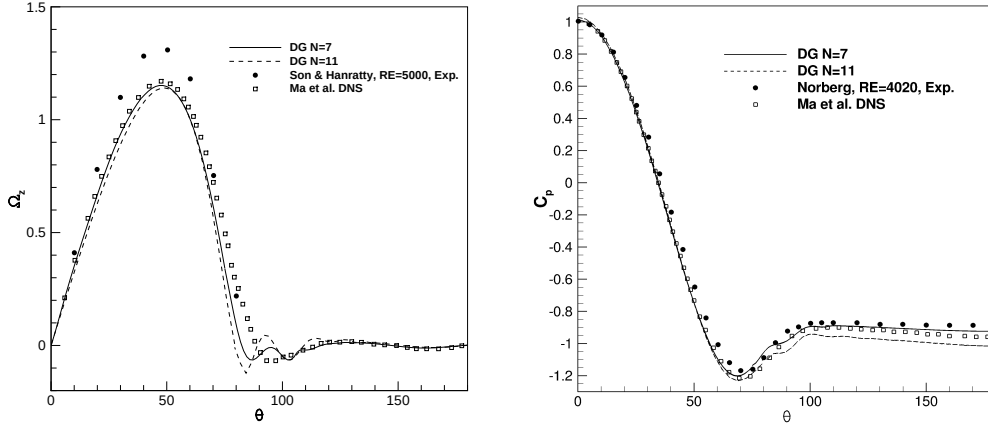


Figure 4. Time and spanwise averaged quantities on the cylinder surface. *left*: Spanwise vorticity component, *right*: Pressure coefficient. Filled circles denote experimental data for $Re = 5000$ of Son & Hanratty for vorticity (taken from [54]) and for $Re = 4020$ of Norberg for pressure coefficient (taken from [48]). Open squares denote DNS results [54]. Solid and dashed lines are current DG results for $N = 7$ and $N = 11$ computations.

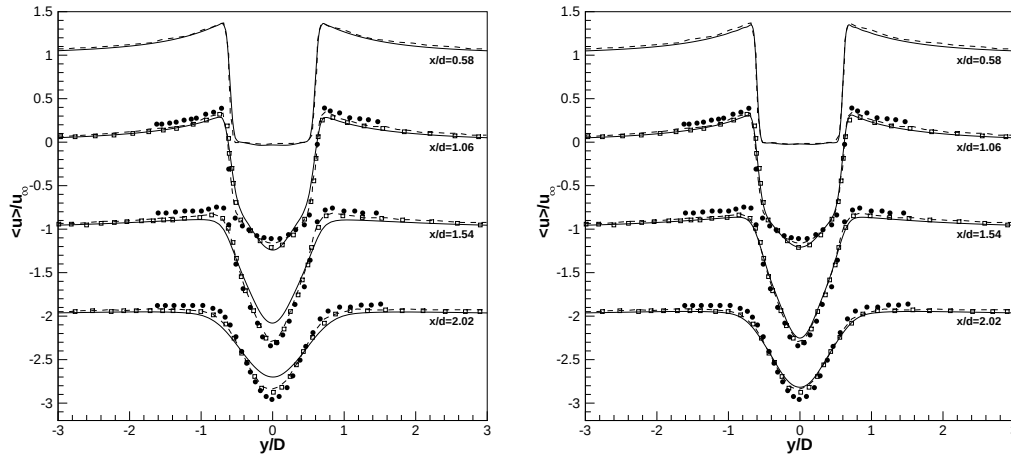


Figure 5. Mean streamwise velocity at different downstream locations in the wake of the circular cylinder flow at $Re_D = 3900$. *left*: $N = 11$ computation, *right*: $N = 7$ computation. Solid lines: present DG results, dashed lines: computational LES results from [58], open squares: computational LES results from [48], filled circles: experimental results from [63].

[58] is very good at all locations. While the $N = 11$ results follow the trend well, their overall match is weaker, as observed for the streamwise components.

3.3. Computational Cost

A meaningful measurement of computational cost to compare different schemes in terms of efficiency is very hard to establish, since implementation details, parallelization strategies, compiler options, computing hardware and network specifications all contribute to the final wall clock time of a given computation. Thus, these numbers are rarely reported in literature. We provide some details here for our cylinder flow computations as a reference.

Tab. III lists the details for our runs. All computations were conducted on the largest number of cores

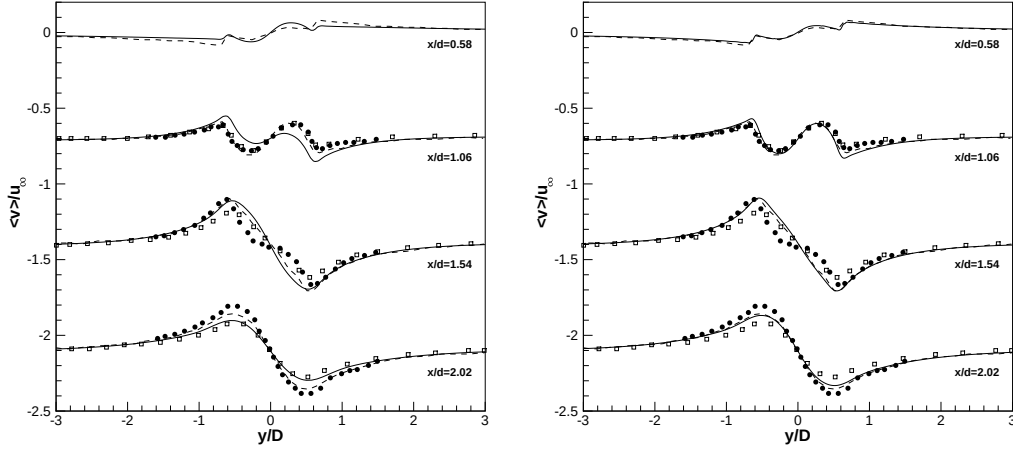


Figure 6. Mean crossflow velocity at different downstream locations in the wake of the circular cylinder flow at $Re_D = 3900$. *left*: $N = 11$ computation, *right*: $N = 7$ computation. Solid lines: present DG results, dashed lines: computational LES results from [58], open squares: computational LES results from [48], filled circles: experimental results from [63].

possible, given by the number of elements in the mesh and the MPI parallelization of our scheme. The average time step $\overline{\Delta t}$ (normalized by the convective time unit T^*) of our fully explicit time integrator is governed by the speed of the sonic wave, due to the low Mach number of 0.1. It should be noted that the time step is indeed larger for the high order scheme. This is counterintuitive, as the time step for advection (which is the restricting time step in these computations) is of the form

$$\Delta t_{max}^{adv} \sim \frac{1}{\lambda_{max}^{adv}} \frac{\Delta x}{2N + 1}, \quad (19)$$

with Δx as the grid spacing and λ_{max}^{adv} the maximum eigenvalue of the flux Jacobian matrix for advection [26]. Therefore, a higher N introduces a stricter time step restriction, but since the grid spacing is larger by a factor of 2 as well, these two factors combine to give a relationship for the time step sizes:

$$\frac{(\Delta t)^{N=7}}{(\Delta t)^{N=11}} \sim \frac{2 \cdot 11 + 1}{2 \cdot 7 + 1} \cdot \frac{(\Delta x)^{N=7}}{(\Delta x)^{N=11}} = \frac{23}{15} \cdot \frac{1}{2} = 0.7667 \quad (20)$$

This estimation for the time step ratio is very close to the actually measured one of 0.7647, computed from the data in Tbl. III.

To facilitate comparisons between different computing architectures, we also compute the performance index PID as

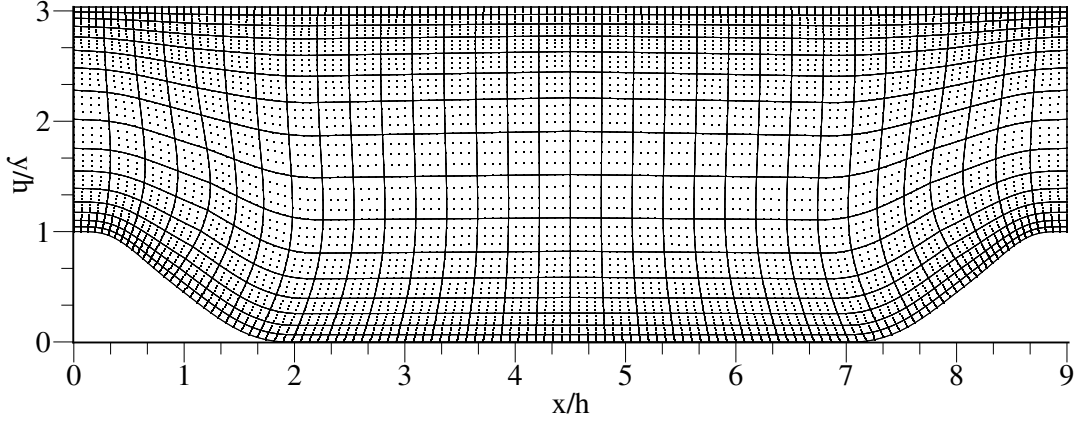
$$PID := \frac{T_{CPU}}{n_{DOF} \times n_{\Delta t}} [s], \quad (21)$$

where $n_{\Delta t}$ denotes the number of explicit time steps and the total CPU time T_{CPU} is computed as the number of computing cores n_{cores} times the simulation wall clock time T_{Run} . The number of spatial degrees of freedom per state vector component n_{DOF} is given by the product of the number of grid cells n_{cells} and the collocation points per element $(N + 1)^3$. While the absolute value of the PID is strongly dependent on the computing systems, its relative changes within a single system allow us to find the optimal configuration in terms of processor load for a given system.

In the cylinder flow case, the PID is better for the $N = 7$ case, as its core load is closer to its

Table III. Computational cost for the cylinder flow computations.

Case	no. DOF	no. cells	cores	$\overline{\Delta t}/T^*$	PID [μ s]	T_{CPU}/T^* [h]	T_{Run}/T^* [min]
$N = 7$	1.572 M	3072	3072	$1.3E - 3$	13	135	2.9
$N = 11$	0.664 M	384	384	$1.7E - 3$	19	105	16.9

Figure 7. Curved mesh with $N_{Geo} = 3$ used for all the periodic hill computations; macro cells: lines; interpolation points: black dots.

optimal value. This is also reflected in the comparison of the total CPU time per convective time unit T_{CPU}/T^* . Although the number of degrees of freedom for $N = 7$ is over twice as high as for $N = 11$, the characteristic total time is only about 25% larger. Tab. III also includes the associated wall clock time per convective unit T_{Run}/T^* in minutes. Due to the efficient parallelization of our scheme, we are able to compute a LES for this case in a matter of hours.

3.4. Discussion

Summarizing the results presented in Sec. 3.2, our de-aliased Discontinuous Galerkin computations show a very good agreement with published data. We employ high order schemes with near spectral cut-off dissipation behavior without LES modelling to compute these flows with a low number of degrees of freedom. The $N = 7$ computations at 1.6 M DOF show excellent matches with experimental results and are able to recover the shape of the near wake almost perfectly. The results of the $N = 11$ computations with 0.7 are also in good agreement with reference data, but show a shorter wake region with associated higher losses. This can be attributed to the overall underprediction of dissipation, due to the very low number of DOF and the higher polynomial ansatz with lower numerical dissipation.

4. PERIODIC HILL CHANNEL FLOW

For this test case we present de-aliased computations in a streamwise and spanwise periodic channel with curved bumps on the lower wall. The geometry of the periodic hill has initially been proposed by Almeida et al. [1], who have experimentally investigated two prototypical model hills in a water

channel, a single hill configuration and a periodic hill configuration.

The periodic hill soon became a frequently used benchmark case for turbulence modelling [57, 68, 32, 15]. In contrast to the more common square duct hill flows, it covers various physical phenomena like flow separation, reattachment and, depending on the Reynolds number, turbulent transition and relaminarization.

When the first steps were undertaken to obtain numerical results, some problems of the experimental configuration were discovered. The high Reynolds number ($Re_h = 60,000$ based on the hill height h) induces high computational cost for a benchmark problem and the influence of the side walls on the flow characteristics is unclear. The latter is problematic, as it is desirable to use periodic boundary conditions in spanwise direction, for both simplicity and to lower computational cost.

Therefore a revised setup has been proposed by Mellen et al. [57], which is more suitable for numerical investigations and soon became more widespread as it was included in the ERCOFTAC database [†]. For the new setup the hill shape remains unchanged, but the distance between the hills is increased, which allows the flow a more natural reattachment process with less interference from the upcoming hill. This increases the variety of flow features and thus makes the case especially demanding in terms of numerical accuracy and modelling aspects.

As experimental results for this configuration were not available yet, first thorough DNS and LES results were provided by Temmerman et al. [68] and subsequently by Fröhlich et al. [32]. Further numerical investigations comparing two FV codes were performed by Breuer et al. [15], who also provided experimental PIV-measurements for the case. Their DNS results cover Reynolds numbers from 100 to 10595 and are also used as an ERCOFTAC reference, where the testcase description and the results are available, that is why we use these results as a benchmark for our computations.

4.1. Computational Setup

4.1.1. Resolution and Discretization The shape of the hill is described in [1], the length of the channel is $L_x = 9h$ and corresponds to the distance between the two hills with h denoting the hill height. The channel height is $L_y = 3.035h$ and the spanwise extent is $L_z = 4.5h$. For the computation we choose a Reynolds number of $Re_h = 2800$ based on the bulk velocity u_b at the hill crest. Our computational mesh is extremely coarse, using 32 elements in streamwise and 16 elements in spanwise and normal direction equaling a total number of only 8192 grid cells. To achieve a meaningful resolution and to follow our assessment of high order schemes, we perform $N = 9$ computations and compare these with $N = 5$ and $N = 3$ computations. We achieve an accurate representation of the geometry by employing a curved mesh, using a polynomial degree of $N_{Geo} = 4$ to represent the curved hill geometry. A high order approximation of the geometry is a requirement to achieve full order of convergence for DG methods, as shown in [7]. We not only use curved elements for boundaries, but for the whole domain, which gives us the chance to apply an inner stretching to the cells, as described in [42]. By doing so, we can distribute the schemes' sub-cell resolution inside the cell to better resolve the boundary layer and improve y^+ . The mesh is depicted in Fig. 7, showing the macro cells as well as the interpolation points.

[†]http://uriah.dedi.melbourne.co.uk/w/index.php/UFR_330

Table IV. Simulation parameters for the $Re_h = 2800$ periodic hill channel flow.

<i>Author</i>	Code	Scheme	Solver	DOF	$\Delta t/10^{-4}$
Breuer et al. [15]	LESOCC	FV	Incompressible	13.1 M	10
Breuer et al. [15]	MGLET	FV-IB	Incompressible	48.0 M	20
Current: $N = 3$	Flexi	DGSEM	Compressible	0.52 M	6.2
Current: $N = 5$	Flexi	DGSEM	Compressible	1.77 M	1.9
Current: $N = 9$	Flexi	DGSEM	Compressible	8.19 M	1.34

Originating from a water channel flow, the periodic hill case has been designed to be incompressible, thus we are again using $Ma = 0.1$ to avoid compressibility effects. As already mentioned above, this testcase would require a $\frac{3}{2}N$ quadrature rule for exact integration. We found that using over-integration by one single additional mode, i.e. $M = N + 1$, is sufficient to stabilize all three computations. As the results presented in Sec. 4.2 indicate, the spatial resolution in our setting is already high, so the non-linear aliasing effects are relatively small. For higher Reynolds numbers or lower resolution, higher quadrature precision is necessary. We use Roe's approximate Riemann solver to compute the fluxes at the boundaries and apply the BR1 method by Bassi and Rebay for the viscous fluxes.

4.1.2. Boundary Conditions and Forcing Term As our domain is periodic in streamwise and spanwise direction, the only boundary conditions we have to consider are the upper and the lower wall, for which we employ isothermal no-slip boundary conditions using a non-dimensional temperature of 1.

However, due to the lack of an inflow boundary, we have to consider a streamwise pressure gradient $\frac{dp}{dx}$ for driving the flow. The pressure gradient is implemented as a forcing term, which we choose to be constant over the whole domain. While it would be possible to prescribe a pressure gradient, which is also constant in time, our goal is to achieve the prescribed mass flow rate $\dot{m}_{ref} = 9.1575$ corresponding to our chosen Reynolds number. In order to control the mass flux $\dot{m}$, we continuously adjust the forcing, which we have implemented by using a backward difference in time

$$\frac{dp^{n+1}}{dx} = \frac{dp^n}{dx} + \frac{1}{(L_y - h)L_z\Delta t}(\dot{m}_{ref} - 2\dot{m}^n + \dot{m}^{n-1}), \quad (22)$$

with n denoting the time step. This procedure has been proposed by Benocci et al. [12], an alternative, more sophisticated method to control the forcing is presented by Lenormand et al. [49], but we found the above method to be sufficiently stable and accurate for small timesteps. Although commonly the mass flux is either measured above the hill crest or (as by Lenormand) along streamwise planes through the domain, we choose a different strategy and use the average mass flux of the whole domain, which to our experience reduces unwanted temporal oscillations of the forcing terms.

4.2. Results

The time-averaged streamlines in Fig. 8 give a general impression of the flow field at $Re_h = 2800$. All time averaged data was collected over 40 convective times $T^* = L_x/u_b$. The flow separates roughly at $x \approx 0.2$ on the lower wall, which is an effect of the high curvature of the descending hill,

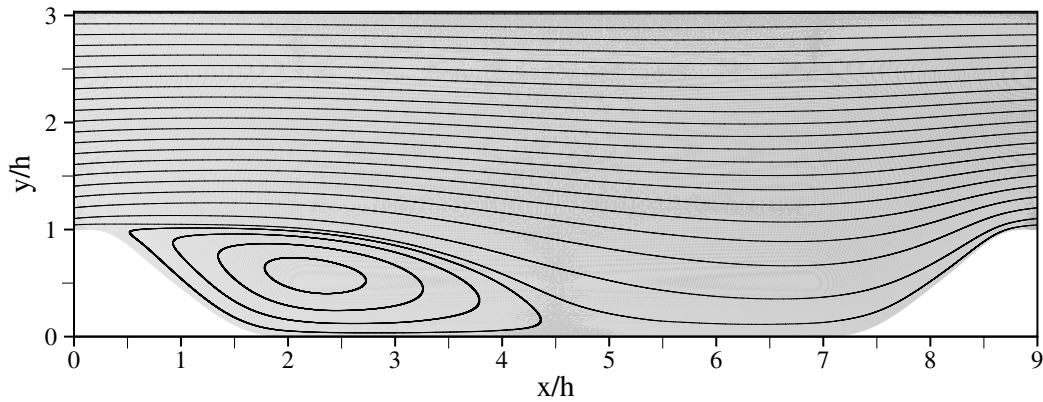
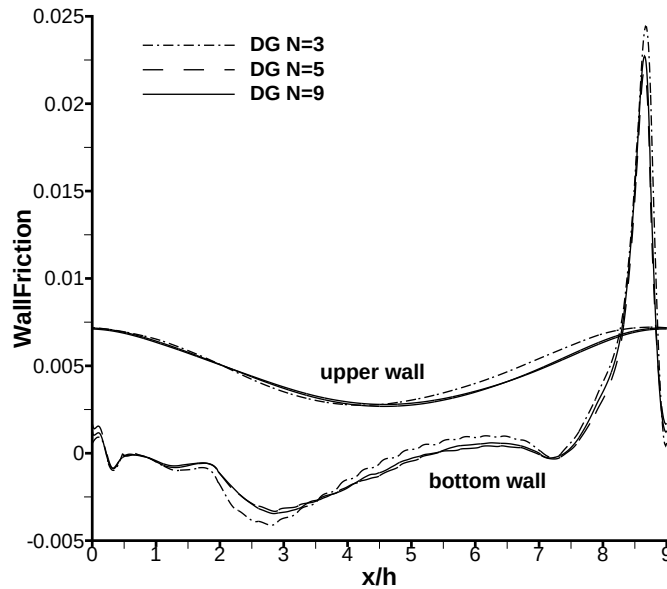


Figure 8. Streamlines of the time and spanwise averaged flow.

Figure 9. Distribution of the time averaged wall friction in streamwise direction on the lower and upper wall, $N = 3$, $N = 5$ and $N = 9$ DGSEM computations.

and reattaches at $x \approx 5$. The recirculation bubble inbetween is known to be strongly dependent on separation effects [68]. Breuer et al. [15] report a small recirculation region at the beginning of the second hill, which could be confirmed by our computations.

The wall friction shown in Fig. 9 converges rapidly with increasing resolution near the walls, as the $N = 5$ and $N = 9$ nearly coincide. As expected, the skin friction rises significantly at the onset of the downstream hill, with its peak on the second plateau, where the biggest friction is expected to occur. The wall friction is negative in the region of the recirculation bubble and defines the size of the region. On the upper wall we see a constant decrease of the friction until the middle of the channel. This corresponds to the decrease of the velocity caused by the widening of the flow field as the normal extent of the recirculation bubble decreases.

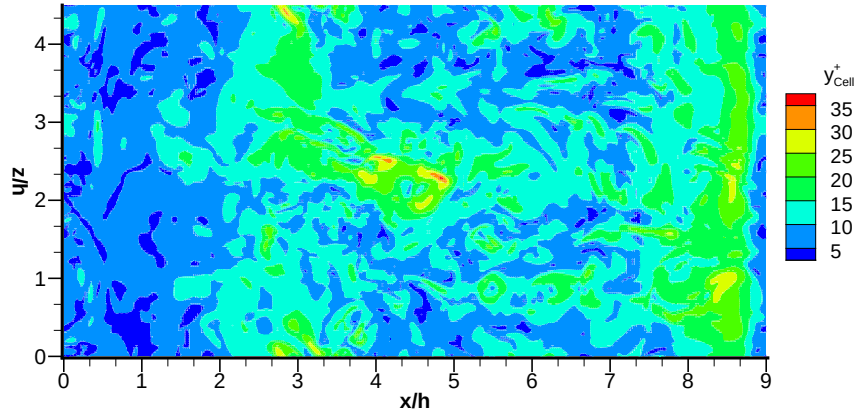


Figure 10. Instantaneous y_{cell}^+ distribution based on the size of the first grid cell on the lower channel wall.

In Fig 10, the instantaneous y_{cell}^+ distribution based on the height of the first grid cell on the lower channel wall is shown. As indicated by the convergence of the skin friction in Fig. 9, the distributions are nearly identical for all the three computations. While at the first sight the y_{cell}^+ values seem very large, one has to consider the subcell resolution of the scheme. A good approximation for the effective y^+ is given by $y^+ \leq y_{cell}^+/(N+1)$. Given the fact that this is a very conservative estimate based on an equispaced distribution of inner cell nodes and that we do not take the aforementioned inner cell stretching into account, our effective y^+ is below or close to the critical limit of $y^+ = 1$.

For a detailed flow analysis, we pick the hill crest and the recirculation zone as a showcase, shown in Figs. 11 and 12. For the time averaged velocities and Reynolds stresses there are no major discrepancies between our simulations and the references. The flow parameters above the crest meet the results better, as the resolution is higher at this point and there are less complex phenomena.

The results indicate that the $N = 3$ computation suffers from a lack of resolution; noticeable differences occur especially at $x = 2.0$. While the streamwise velocity $\langle u \rangle / u_b$ and Reynolds stresses $\langle u' u' \rangle / u_b^2$ are in good agreement to the ERCOFTAC references, there are major differences in the normal velocities and stresses, the later being very sensitive to the amount of resolution. Oscillations near the peak can be observed for all stresses and turbulent kinetic energy. The time averaging interval for $N = 3$ has been extended to 100 convective times to eliminate the risk of the oscillations stemming from a too short averaging period. We thus suspect that the oscillations indicate slight instabilities. Nevertheless the overall impression is still good, considering barely 0.5 M degrees of freedom have been used.

Both the $N = 5$ and the $N = 9$ computations show in general a very good agreement with the references. While the $N = 5$ tends to have slightly lower peaks than the reference, e.g. in the normal components of the stresses $\langle v' v' \rangle / u_b^2$, the $N = 9$ slightly overshoots the published data. We note that while all three cases do well for the streamwise velocities, the improvements due to additional accuracy become apparent for the spanwise components and stresses.

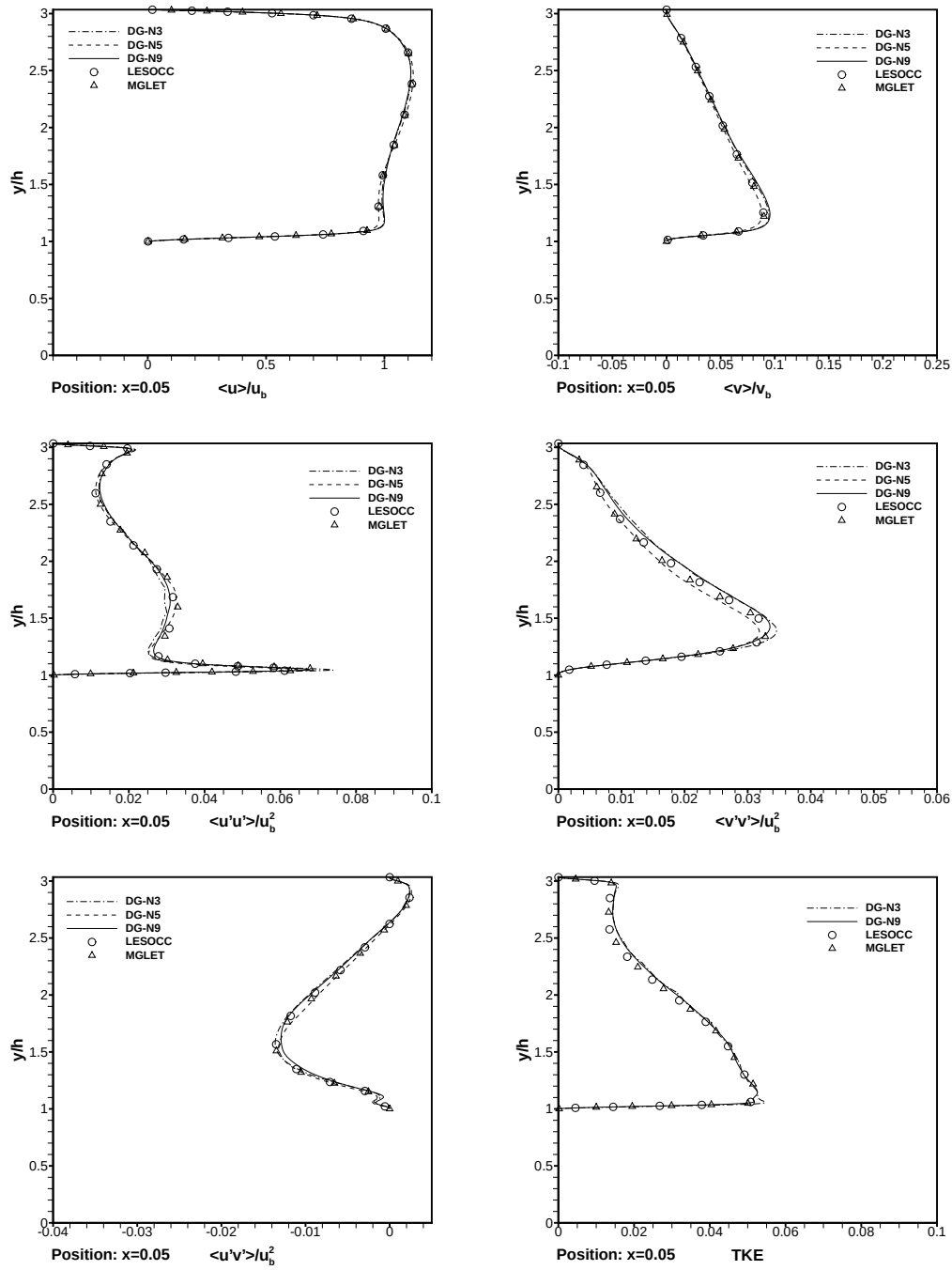


Figure 11. Spatial and time averaged streamwise velocity $\langle u \rangle / u_b$, normal velocity $\langle v \rangle / u_b$, spanwise velocity, the Reynolds stresses $\langle u'u' \rangle / u_b^2$, $\langle v'v' \rangle / u_b^2$, $\langle u'v' \rangle / u_b^2$ and the turbulent kinetic energy at the hill crest ($x = 0.05$).

4.3. Computational Cost

For the periodic hill testcase we achieve a PID in the region $1.0E - 5$. The drop of the PID between the $N = 3$ and $N = 5$ run demonstrates the positive effects of an optimal load per core. The $N = 9$

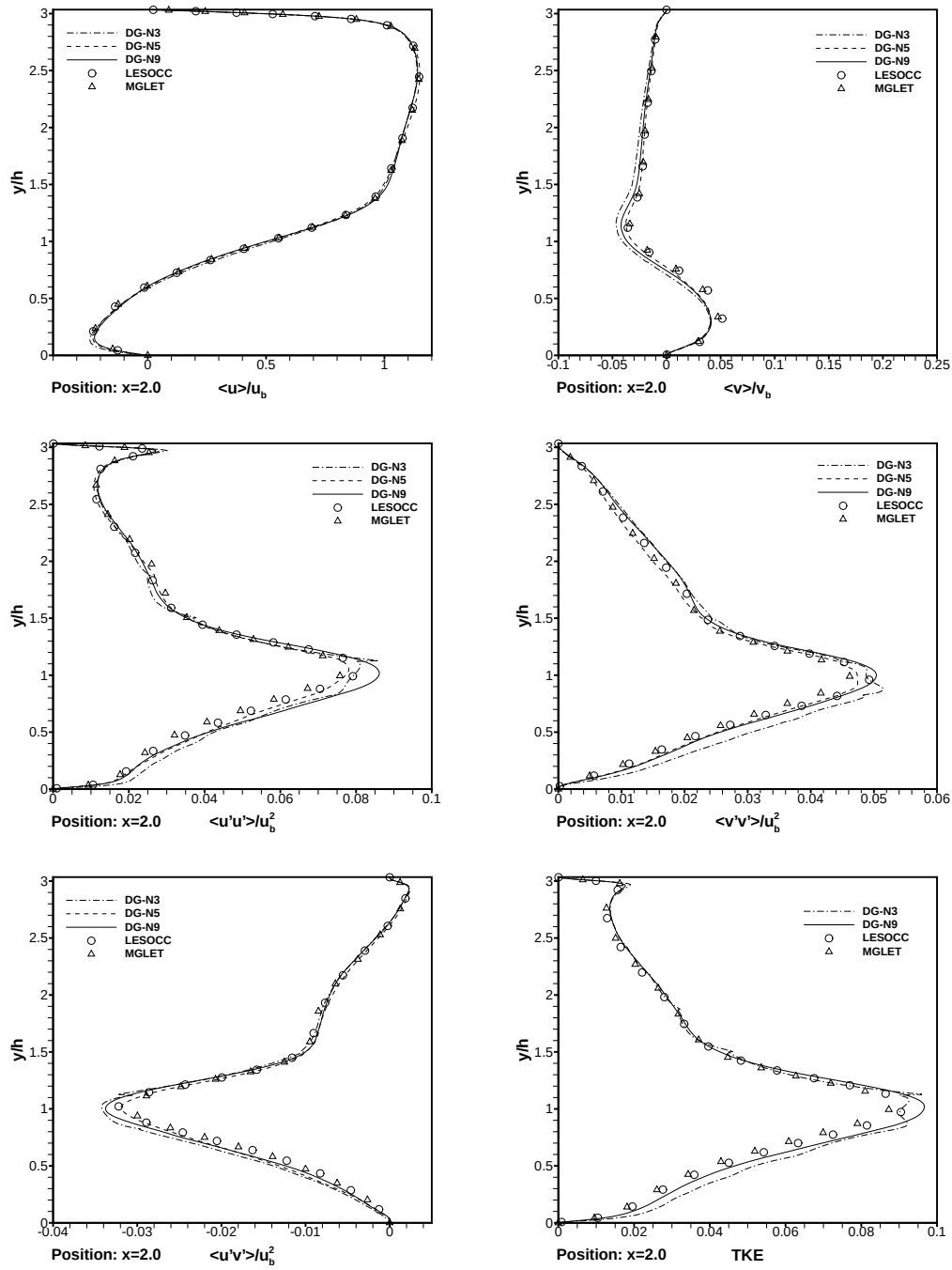


Figure 12. Spatial and time averaged streamwise velocity $\langle u \rangle / u_b$, normal velocity $\langle v \rangle / u_b$, spanwise velocity, the Reynolds stresses $\langle u'u' \rangle / u_b^2$, $\langle v'v' \rangle / u_b^2$, $\langle u'v' \rangle / u_b^2$ and the turbulent kinetic energy in the separation region ($x = 2.0$).

case shows that we are able to go down to a number as low as two elements per core even for a high number of processors and still achieve a feasible PID.

Table V. Computational cost for the periodic hill computations.

Case	no. DOF	cores	no. cells/core	PID [μ s]	$T^*/\Delta t$	T_{CPU}/T^* [h]	T_{run}/T^* [min]
$N = 3$	0.52 M	1024	8	13.9	20161	63	3.7
$N = 5$	1.77 M	1024	8	7.1	65789	287	16.8
$N = 9$	8.19 M	4096	2	14.2	96153	3287	48.0

4.4. Discussion

Summarizing the results of the three periodic hill computations, we show that they are in very good agreement with the reference results. We demonstrate that by using de-aliased high order schemes, one can achieve accurate results for this case with magnitudes less degrees of freedom. A good example is the $N = 3$ computation with 0.5 M degrees of freedom. While it has a factor of 25, respectively a factor of 90 less degrees of freedom than the state-of-the-art FV codes LESOCC and MGLET the results are still in close accordance with the published data. On the other hand, the $N = 5$ and $N = 9$ computations match very well with the reference results with negligible differences while still having considerably less degrees of freedom.

These results indicate that even with an incomplete de-aliasing strategy good results can be achieved with a well-balanced ratio between resolution and integration accuracy.

5. DETACHED AIRFOIL FLOW SIMULATIONS

The prediction of flows at low to medium Reynolds numbers (10^4 to 10^6) gains rising attention in applications such as micro-aerial and unmanned vehicles. As these flows exhibit substantial laminar regions, a laminar separation bubble (LSB) is likely to form even at modest angles of attack, followed by transition in the separated shear layer and subsequent reattachment. The concurrent, coupled occurrence of those three different flow regimes makes this type of flow very hard to predict with Reynolds Averaged Navier Stokes simulations. On the other hand, the computational cost of direct numerical simulations in this Reynolds number regime is still very high. This leaves LES as the main candidate for the prediction of this type of flow.

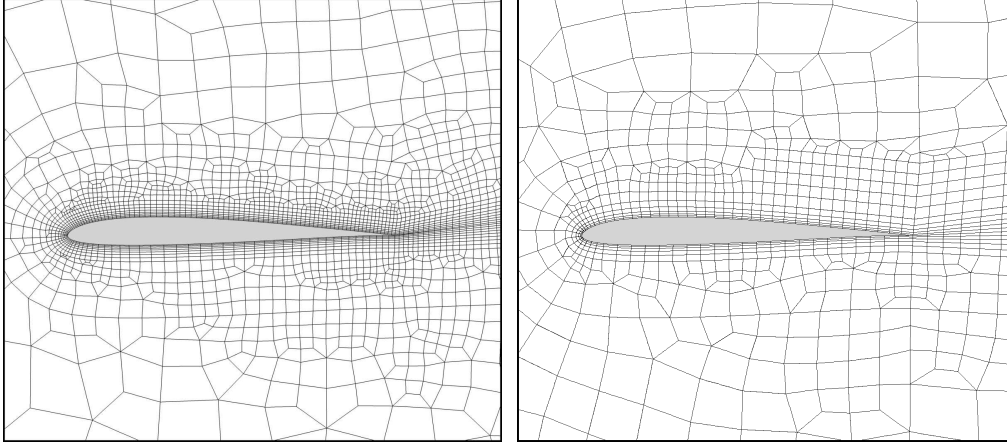
We present the simulation around a Selig-Donavan (SD) 7003 airfoil at $Re_c = 60,000$ and angle of attack of $\alpha = 8^\circ$. This airfoil has been subject of experimental [60, 66] and some computational investigations in the last years [34, 21, 14], most of them using implicit or explicit LES approaches. At present, no fully resolved DNS computation is available for reference. This setup features a LSB at the leading edge, a rapidly growing Kelvin-Helmholtz instability leading to transition to turbulence and subsequent reattachment at approximately the first third of the chord.

5.1. Computational Setup

Following the instructions from the first [71] and second “International Workshop on High Order CFD-Methods”, we run this case at $Ma = 0.1$ and employ isothermal walls with $T_{wall}/T_\infty = 1.002$. For this case, the Roe Riemann solver and the BR1 scheme for the viscous fluxes is employed.

Table VI. Degrees of freedom and resolution details of the SD7003 airfoil flow simulations.

Case	DOF	no. elements	no. elements spanwise	$y_{x=0.8}^+$	N	M
$N = 3$	4.26 M	66,500	12	1.65	3	4
$N = 7$	4.55 M	8,900	6	1.10	7	9

Figure 13. Computational mesh for the $N = 3$ (left) and the $N = 7$ case (right).

The computational domain is a circle of 100 chords radius, in order to minimize the influence of the freestream boundary conditions on the aerodynamic near field. We discretize this domain with unstructured hexahedra, which allows a very efficient and coarse resolution of the farfield. To account for the airfoil curvature, the geometry is represented through a mapping with $N_{Geo} = 4$ within the wall-adjacent cells. All other cells remain with a trilinear mapping. At the farfield and outflow boundaries, weakly imposed Dirichlet boundary conditions by means of a numerical flux are used. At the outflow, the continuous grid coarsening effectively damps the turbulent wake towards the boundary, preventing spurious reflections. The spanwise extent is 0.2 chord lengths C and periodic boundary conditions are employed at the spanwise boundaries. Two different simulations with polynomial degrees $N = 3$ and $N = 7$ are performed on two different meshes, keeping the number of degrees of freedom approximately constant. As can be observed in Fig. 13, the resulting mesh for the $N = 7$ case is significantly coarser. Details on the two calculations, including the wall-normal resolution in wall coordinates at $x = 0.8$ (coordinate origin at the leading edge) are listed in Tab. VI. Note that y^+ is normalized by the number of cell-internal degrees of freedom in wall-normal direction, i.e. $y^+ = y_{cell}^+ / (N + 1)$. Both simulations are initialized with the freestream state. We start to collect statistics after 20 convective times (C/U_∞), the time-averaging interval is $10 C/U_\infty$.

5.2. Results

Fig. 14 illustrates the main flow features by means of contour plots of the velocity parallel to the chord $\langle v_1 \rangle$ and the respective fluctuation energy $k = \langle v'_i v'_i \rangle$. Due to the relatively high angle of attack, the boundary layer separates in direct vicinity of the leading edge and a LSB forms. From the time averaged flow field, a large recirculation vortex is observable. The fluctuation energy contours

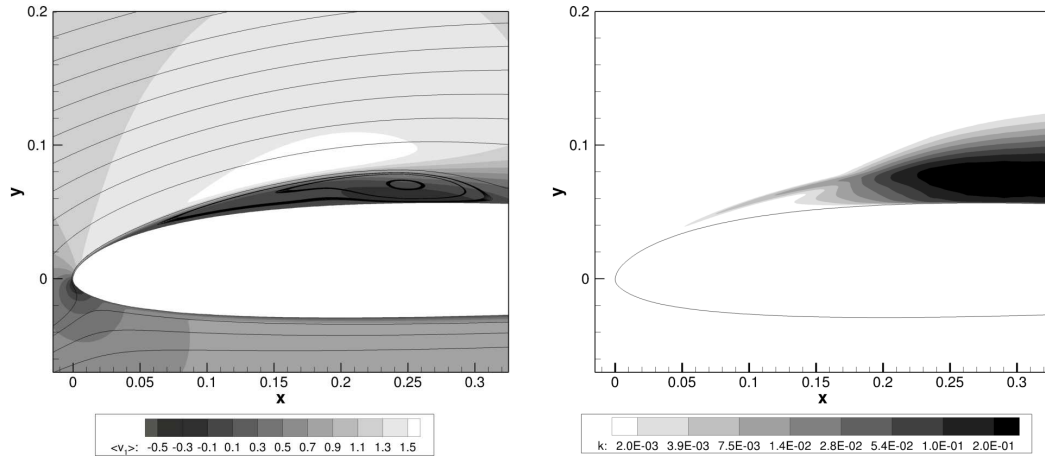


Figure 14. Time-averaged velocity parallel to the chord $\langle v_1 \rangle$ with streamlines (left) and fluctuation energy k (right) for the $N = 7$ case.

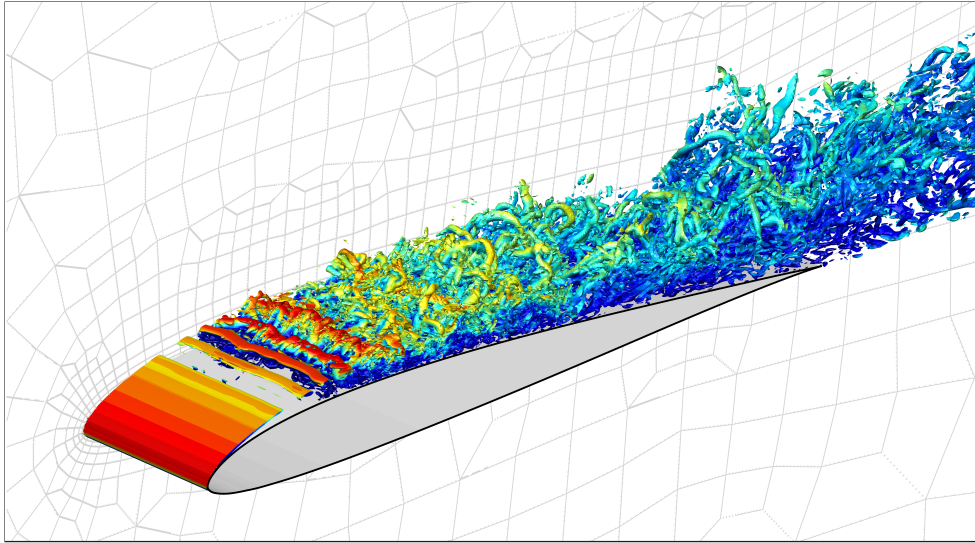


Figure 15. Isosurfaces of the Q criterion ($Q = 500$) colored with contours of instantaneous velocity magnitude.

are distributed exponentially in order to visualize the disturbance growth within the separated shear layer. Additionally, redistribution of turbulent material underneath the shear layer towards the leading edge is observable. From the vortex structures visualized in Fig. 15 by means of isosurfaces of the Q criterion, the build-up of spanwise Kelvin-Helmholtz vortices and subsequent secondary instabilities are visible, as well as the immediate breakdown to turbulence.

The aerodynamic loads as well as separation and reattachment positions predicted by our computations can be compared to other published computational results and the experimental data of Selig et al [66] in Tab VII. The current results, in particular the $N = 3$ case agree well with the results of other groups. Compared to the results of Galbraith and Visbal [34] and Boom and Zingg [14], separation occurs earlier and reattachment later, a trend which intensifies for the higher order

Table VII. Mean aerodynamic loads and separation (x_s) and reattachment (x_r) locations and computational details of the SD7003 airfoil case.

Author	C_L	C_D	x_s	x_r	Scheme	DOF
Galbraith & Visbal [34]	≈ 0.91	≈ 0.043	0.040	0.280	$\mathcal{O}(6)$ Comp. FD	5.70 M
Catalano & Tognaccini [21]	≈ 0.94	≈ 0.044	0.030	0.290	$\mathcal{O}(2)$ FD & dyn. Smag.	8.63 M
Boom & Zingg [14]	0.968	0.034	0.037	0.200	$\mathcal{O}(4)$ SBP-SAT	4.48 M
Selig et al (exp.) [66]	≈ 0.92	≈ 0.029	—	—	—	—
Current: $N = 3$	0.923	0.045	0.027	0.310	DGSEM	4.26 M
Current: $N = 7$	0.932	0.050	0.030	0.336	DGSEM	4.55 M

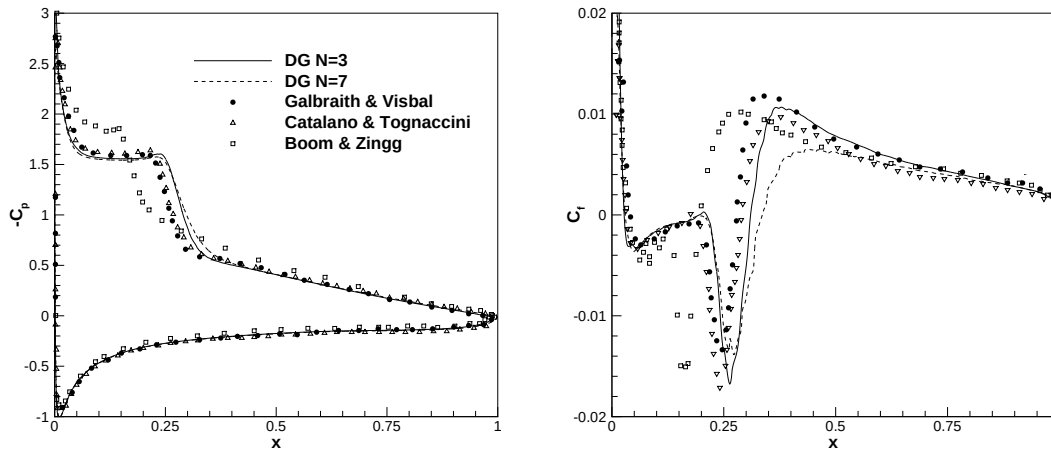


Figure 16. Pressure coefficient for suction and pressure side (left) and skin friction coefficient for the suction side (right), SD7003 airfoil.

case $N = 7$. While the experimental lift coefficient is predicted reasonably, the drag is overestimated by all computations.

The pressure coefficient distributions are shown in Fig. 16, demonstrating fair agreement with all computations but the one of Boom and Zingg, where the LSB is shifted towards the leading edge. For the remainder of the references, good agreement is also found for the skin friction distribution until $x \approx 0.2$ and from $x \approx 0.5$. From the skin friction coefficient distribution, it becomes evident that the recirculation and reattachment area is critical. Discrepancies in the shape and size of the footprint of the recirculation vortex are present among all available datasets. The considerable spread particularly in the skin friction coefficient distributions may be explained to some extent by the different domain sizes. For example, the farfield boundary in the simulation of Boom and Zingg was placed at 15 chords. In our simulations, a strong influence of the farfield boundary was observed: we found noticeable differences in the length and height of the separation bubble between simulations with a 20 and a 50 chords computational domain. Yet this does not explain the deviation of the skin friction in the reattachment region between the $N = 3$ and $N = 7$ cases. Reasons may be local insufficient resolution or the different numerical dissipation characteristics. The above considerations underline the general need for DNS reference data for this case.

In order to assess the turbulent energy content, we extract the time data with virtual probes and perform FFT of the time signal to obtain spectra of kinetic energy. Assuming mere advection

of the turbulent fluctuations with the time-averaged velocity $\langle \vec{v} \rangle$, it is possible to estimate the corresponding wave number as

$$k = \frac{2\pi f}{\langle \|\vec{v}\| \rangle}, \quad (23)$$

which is known as Taylor's hypothesis [67] and very popular in laboratory experiments. Taylor reasoned (23) should be a good approximation when the turbulence intensity u' is small compared to the mean flow $\langle u \rangle$. Recently, del Alamo and Jiménez [30] analyzed the approximation for channel flow and found that while small scale fluctuations are transported with the local mean velocity close to the wall, for larger long-wavelength structures the advection velocity is close to the bulk velocity. Despite these shortcomings, we will use this approximation as a means to analyze the spectral energy content and compare it with the numerical approximation properties of the scheme.

Fig. 17 shows an instantaneous velocity field and pseudo spatial spectra obtained through (23). Temporal samples are taken at every fifth time step over ten convective times C/U_∞ , summing up to a total of about 270,000 samples. The interval is subdivided into five blocks, over which the spectra are averaged. The local time-averaged velocity magnitude is used for each probe to evaluate (23). At the location of probe 1 (Fig. 17 left), the prerequisites for Taylor's hypothesis are fulfilled and we find a well developed inertial range in good agreement with the theoretical $k^{-5/3}$ law, followed by the spectral decay towards higher wavenumbers. The dissipation rate obtained by

$$\epsilon(k) = 2\nu k^2 E(k) \quad (24)$$

is plotted in the same diagram. From the local cell size in direction of the mean advection velocity, we calculate the limits of the local marginal wave number region between 2 and 4 points per wavelength (PPW) or $\pi/2 \leq k^* < \pi$ from

$$k^* = \frac{k\Delta x}{N+1}. \quad (25)$$

Thereby we exploit the fact that the advection velocity is – in good approximation – aligned with the local cell for our three probes. Let us first take a closer look at the high wavenumber limit ($k^* = \pi$) for probe 1. Despite the averaging, the spectra still features small spikes. However, very close to $k^* = \pi$, the spectrum transitions to a very smooth continuous shape. It seems natural that the point of this transition marks the absolute limit of the spatial representation. The almost perfect coincidence of the Nyquist wavenumber calculated from (25) and this transition point strengthens the validity of Taylor's hypothesis for probe 1.

The left limit of the gray shaded band indicates that the lower wavenumber range to the left is well-resolved. Moreover, the peak dissipation is within the well-resolved wave number region. From the theoretical approximation properties of our scheme, in particular for high polynomial degrees, we expect that most of the numerical dissipation is active in the band $\pi/2 \leq k^* < \pi$. Probe 2 has been chosen as an example where Taylor's hypothesis fails due to a very high fluctuation to advection velocity ratio and high mean shear close to the wall. The transition to the smooth spectrum occurs at more than twice the nyquist wavenumber $k^* = \pi$. As the $k^* = \pi$ value is a direct result of the local cell size, the spectra would need to smoothen out around that point for a valid Taylor's hypothesis. Probe 3 lies in a slightly less resolved region, so the gray region is shifted to lower wavenumbers compared to probe 1 and 2. The spectral behavior of probe 1 has been confirmed for

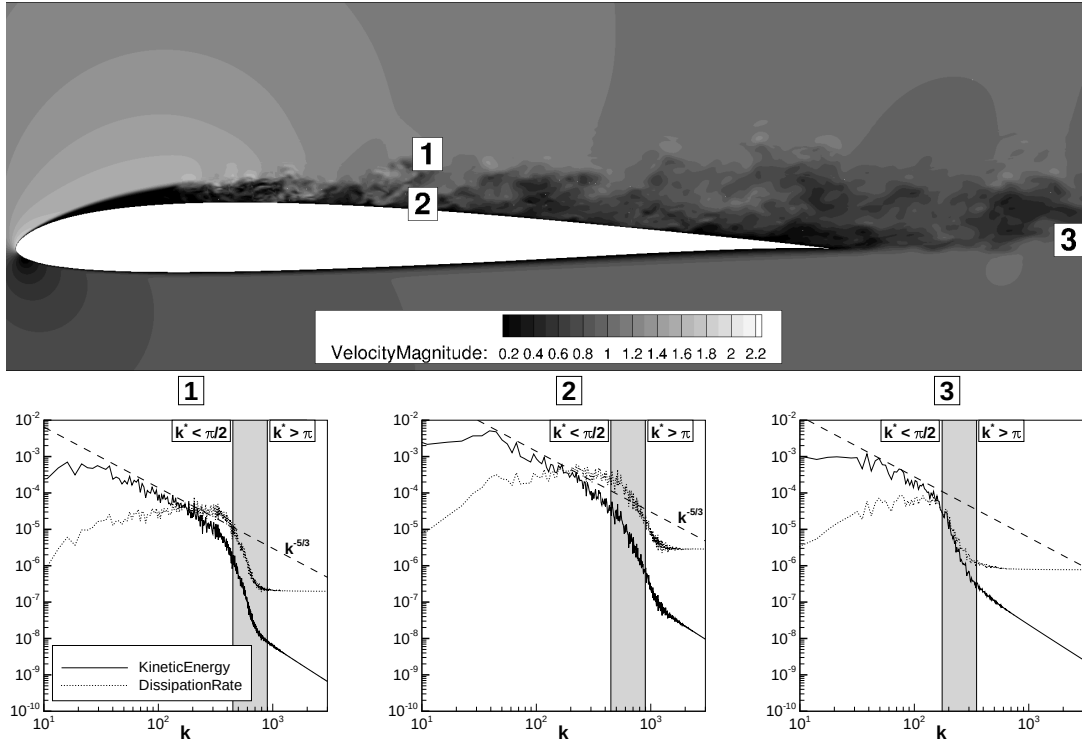


Figure 17. Top: Instantaneous velocity magnitude contours at $z = 0.1$. Numbers mark the corresponding spectra. Bottom: Pseudo-spatial kinetic energy and dissipation spectra at the marked positions (left: $(x, y, z) = (0.5, 0.11, 0)$, center: $(x, y, z) = (0.5, 0.05, 0)$, right: $(x, y, z) = (1.3, 0, 0)$).

Table VIII. Computational cost for the SD7003 airfoil computations.

Case	no. DOF	cores	$\overline{\Delta t}/T^*$	no. cells/core	PID [μ s]	T_{CPU}/T^* [h]	T_{Run}/T^* [min]
$N = 3$	4.26 M	5548	$9.7E - 6$	12	13.4	3200	34.8
$N = 7$	4.55 M	4443	$7.3E - 6$	2	12.5	4220	57

more points within the turbulent boundary layer downstream of the reattachment point, as long as they were sufficiently far away from the mean shear region close to the wall.

5.3. Computational Cost

Details on the computational cost of the two cases are listed in Tab VIII. Interestingly, the PID for the $N = 7$ case is even lower than for the $N = 3$ case, showing that the high order operator is not necessarily more expensive. The higher CPU time stems mostly from a slightly lower time step.

5.4. Discussion

The transitional SD7003 airfoil flow computations show mostly good agreement with published results in integral quantities and pressure and skin friction coefficient distributions. However, especially in the skin friction coefficient, spread among the available references and our results is present. For the high angle of attack of $\alpha = 8^\circ$, the LSB is relatively short and the complex transition and reattachment processes occur on a very small portion of the airfoil. The distance to

the farfield boundaries strongly affects the properties of the LSB, which presumably causes part of the spread. The complex physics renders this setup a good LES test case, making DNS reference data desirable. The turbulence spectra analysis using Taylor's hypothesis is presented as a means of assessing the local resolution of turbulence.

6. CONCLUSION

In this paper, we have investigated the accuracy and efficiency of a high order Discontinuous Galerkin discretization for the simulation of under-resolved transitional and turbulent flows. While a number of simulations with moderate order DG schemes with implicit time integration for the solution of the RANS equations have been reported in literature, the efforts to extend the applicability to LES are not as pronounced. This is remarkable, as our results indicate that especially LES computations can benefit greatly from the near-spectral approximation properties and very high resolution per degree of freedom of high order approximations.

Within this work, we have applied high order schemes to three classical test cases for the evaluation of LES capabilities. We use de-aliasing of the non-linear terms to avoid aliasing and to recover the fundamental properties of the DG formulation. The total number of degrees of freedom for our DG solution was taken to be equal or considerably less than reported in the references. We employ no explicit or implicit LES modelling, i.e. we add no modelling terms or filters to the equation, and we do not adapt the discretization to account for the unresolved subgrid effects. Instead we rely on the high resolution quality per degree of freedom of high order polynomial approximations and the cut-off like dissipation at marginally resolved scales of the DG operator. These two features allow a high fidelity resolution of a broad range of scales, which introduce the associated dissipation in a consistent way and produce an efficient energy destruction at the resolution limit.

With this strategy, we have shown that a high order standard DG scheme without additional LES modelling is capable of matching or surpassing the solution accuracy of established LES schemes for the simulation of medium Reynolds number flows. We have also demonstrated that we are able to compute these flows in wall clock times on the order of a few hours.

It is also important to remark that we acknowledge the limits of this approach: for higher Reynolds number and less resolution, the inherent numerical dissipation, acting as a sink for the cascading energy, will not be sufficient. Then, high order DG methods will have to be supplemented by a suitable LES modelling strategy for the subgrid terms.

For explicit LES modelling, the spectral cut-off character and the de-aliasing are again beneficial: They guarantee low dissipation and dispersion errors and avoid aliasing, thereby minimizing the overall approximation error. The advantage of our approach is that it ensures minimal interference between numerical errors and the subgrid scale model, thereby allowing a meaningful evaluation and application of these models. Therefore, our future strategy will be to combine high order de-aliased DG schemes with explicit subgrid scale models.

REFERENCES

1. ALMEIDA, G., DURO, D., AND HEITOR, M. Wake flows behind two-dimensional model hills. *Experimental Thermal and Fluid Science* 7, 1 (1993), 87 – 101.
2. ARNOLD, D. N., BREZZI, F., COCKBURN, B., AND MARINI, L. D. Discontinuous Galerkin methods for elliptic problems. In *Discontinuous Galerkin Methods. Lecture Notes in Computational Science and Engineering*, B. Cockburn, G. Karniadakis, and C.-W. Shu, Eds. Springer, 2000, pp. 89–101.
3. ARNOLD, D. N., BREZZI, F., COCKBURN, B., AND MARINI, L. D. Unified analysis of discontinuous Galerkin methods for elliptic problems. *SIAM J. Numer. Anal.* 39, 5 (2002), 1749–1779.
4. BARTER, G. E., AND DARMOFAL, D. L. Shock capturing with pde-based artificial viscosity for dgfem: Part 1. formulation. *Journal of Computational Physics* 229, 5 (2010), 1810 – 1827.
5. BASSI, F., CRIVELLINI, A., REBAY, S., AND SAVINI, M. Discontinuous Galerkin solution of the Reynolds-averaged NavierStokes and $k\omega$ turbulence model equations. *Computers and Fluids* 34, 45 (2005), 507 – 540.
6. BASSI, F., AND REBAY, S. A high order accurate discontinuous finite element method for the numerical solution of the compressible Navier–Stokes equations. *Jornal of Computational Physics* 131 (1997), 267–279.
7. BASSI, F., AND REBAY, S. A high-order discontinuous Galerkin finite element method solution of the 2d euler equations. *J. Comput. Phys.* 138 (1997), 251–285.
8. BASSI, F., AND REBAY, S. Numerical evaluation of two discontinuous Galerkin methods for the compressible Navier–Stokes equations. *International Journal for Numerical Methods in Fluids* 40 (2002), 197–207.
9. BASSI, F., AND REBAY, S. Numerical evaluation of two discontinuous Galerkin methods for the compressible NavierStokes equations. *International Journal for Numerical Methods in Fluids* 40, 1-2 (2002), 197–207.
10. BASSI, F., REBAY, S., MARIOTTI, G., PEDINOTTI, S., AND SAVINI, M. A high-order accurate discontinuous finite element method for inviscid an viscous turbomachinery flows. In *Proceedings of 2nd European Conference on Turbomachinery, Fluid and Thermodynamics* (Technologisch Instituut, Antwerpen, Belgium, 1997), R. Decuyper and G. Dibelius, Eds., pp. 99–108.
11. BECK, A., GASSNER, G., AND MUNZ, C.-D. On the effect of flux functions in discontinuous Galerkin simulations of underresolved turbulence. In *Spectral and High Order Methods for Partial Differential Equations - ICOSAHOM 2012*, M. Azaez, H. El Fekih, and J. S. Hesthaven, Eds., vol. 95 of *Lecture Notes in Computational Science and Engineering*. Springer International Publishing, 2014, pp. 145–155.
12. BENOCCI, C., AND PINELLI, A. *Engineering Turbulence Modeling and Experiments*. Elsevier, 1990, ch. The role of the forcing term in the large eddy simulation of equilibrium channel flow, pp. 287–296.
13. BLACKBURN, H. M., AND SCHMIDT, S. Large eddy simulation of flow past a circular cylinder. In *Proceedings of 14th Australasian Fluid Mechanics Conference* (2001).
14. BOOM, P. D., AND ZINGG, D. W. Time-accurate flow simulations using an efficient Newton-Krylov-Schur approach with high-order temporal and spatial discretization.
15. BREUER, M., PELLER, N., RAPP, C., AND MANHART, M. Flow over periodic hills numerical and experimental study in a wide range of Reynolds numbers. *Computers and Fluids* 38, 2 (2009), 433 – 457.
16. BURGESS, N., AND MAVRIPLIS, D. J. High-order discontinuous Galerkin methods for turbulent high-lift flows. In *Proceedings of Seventh International Conference on Computational Fluid Dynamics (ICCFD7)*, 2012.
17. CARPENTER, M., AND KENNEDY, C. Fourth-order 2N-storage Runge-Kutta schemes. Tech. Rep. NASA TM 109111, 1994.
18. CARTON DE WIART, C., AND HILLEWAERT, K. DNS and ILES of transitional flows around a SD7003 using a high order discontinuous Galerkin method. In *Seventh International Conference on Computational Fluid Dynamics (ICCFD7)* (2012).
19. CARTON DE WIART, C., HILLEWAERT, K., DUPONCHEEL, M., AND WINCKELMANS, G. Assessment of a discontinuous Galerkin method for the simulation of vortical flows at high Reynolds number. *International Journal for Numerical Methods in Fluids* (2013).
20. CASONI, E., PERAIRE, J., AND HUERTA, A. One-dimensional shock-capturing for high-order discontinuous Galerkin methods. *International Journal for Numerical Methods in Fluids* 71, 6 (2013), 737–755.
21. CATALANO, P., AND TOGNACCINI, R. Large Eddy Simulations of the Flow around the SD7003 airfoil. In *AIMETA Conference* (2011).
22. COCKBURN, B., HOU, S., AND SHU, C. W. The Runge-Kutta local projection discontinuous Galerkin finite element method for conservation laws IV: The multidimensional case. *Math. Comput.* 54 (1990), 545–581.
23. COCKBURN, B., LIN, S. Y., AND SHU, C. TVB Runge-Kutta local projection discontinuous Galerkin finite element method for conservation laws III: One dimensional systems. *J. Comput. Phys.* 84 (1989), 90–113.
24. COCKBURN, B., AND SHU, C. W. TVB Runge-Kutta local projection discontinuous Galerkin finite element method for conservation laws II: General framework. *Math. Comput.* 52 (1989), 411–435.

25. COCKBURN, B., AND SHU, C.-W. The runge-kutta local projection p1-discontinuous Galerkin finite element method for scalar conservation laws. *RAIRO Modél. Math. Anal. Numér* 25, 3 (1991), 337–361.
26. COCKBURN, B., AND SHU, C.-W. Rungekutta discontinuous galerkin methods for convection-dominated problems. *Journal of Scientific Computing* 16, 3 (2001), 173–261.
27. COLLIS, S. Discontinuous Galerkin methods for turbulence simulation. In *In Proceedings of the 2002 Center for Turbulence Research Summer Program* (2002), pp. 155–167.
28. COLLIS, S. The DG/VMS method for unified turbulence simulation. In *32nd AIAA Fluid Dynamics Conference and Exhibit* (2002).
29. CRIVELLINI, A., D’ALESSANDRO, V., AND BASSI, F. High-order discontinuous Galerkin rans solutions of the incompressible flow over a delta wing. *Computers and Fluids* 88, 0 (2013), 663 – 677.
30. DEL ÁLAMO, J. C., AND JIMÉNEZ, J. Estimation of turbulent convection velocities and corrections to taylor’s approximation. *Journal of Fluid Mechanics* 640 (2009), 5.
31. FRANKE, J., AND FRANK, W. Large eddy simulation of the flow past a circular cylinder at $Re_D = 3900$. *Journal of Wind Engineering and Industrial Aerodynamics* 90, 10 (2002), 1191 – 1206.
32. FRÖHLICH, J., MELLEN, C. P., RODI, W., TEMMERMAN, L., AND LESCHZINER, M. A. Highly resolved large-eddy simulation of separated flow in a channel with streamwise periodic constrictions. *Journal of Fluid Mechanics* 526 (3 2005), 19–66.
33. FRÖHLICH, J., RODI, W., KESSLER, P., PARPAIS, S., BERTOGLIO, J., AND LAURENCE, D. Large eddy simulation of flow around circular cylinders on structured and unstructured grids. In *Numerical Flow Simulation I*, E. Hirschel, Ed., vol. 66 of *Notes on Numerical Fluid Mechanics (NNFM)*. Springer Berlin Heidelberg, 1998, pp. 319–338.
34. GALBRAITH, M., AND VISBAL, M. Implicit large eddy simulation of low Reynolds number flow past the SD7003 airfoil. *AIAA paper* 225 (2008), 2008.
35. GASSNER, G. A skewsymmetric nodal discontinuous Galerkin method for the compressible Euler equations. *submitted to International Journal on Numerical Methods for Fluid Dynamics*.
36. GASSNER, G. A skew-symmetric discontinuous Galerkin spectral element discretization and its relation to sbp-sat finite difference methods. *SIAM Journal on Scientific Computing* 35, 3 (2013), A1233–A1253.
37. GASSNER, G., AND BECK, A. On the accuracy of high-order discretizations for underresolved turbulence simulations. *Theoretical and Computational Fluid Dynamics* 27, 3-4 (2013), 221–237.
38. GASSNER, G., AND KOPRIVA, D. A comparison of the dispersion and dissipation errors of Gauss and Gauss-Lobatto discontinuous Galerkin spectral element methods. *SIAM J. Scientific Computing* 33, 5 (2011), 2560–2579.
39. GASSNER, G., LÖRCHER, F., AND MUNZ, C.-D. A contribution to the construction of diffusion fluxes for finite volume and discontinuous Galerkin schemes. *J. Comput. Phys.* 224, 2 (2007), 1049–1063.
40. GASSNER, G., LÖRCHER, F., AND MUNZ, C.-D. A discontinuous Galerkin scheme based on a space-time expansion. II. Viscous flow equations in multi dimensions. *J. Sci. Comp.* 34, 3 (2008), 260–286.
41. HINDENLANG, F., GASSNER, G., BOLEMAN, T., AND MUNZ, C.-D. Unstructured high order grids and their application in discontinuous Galerkin methods. In *Proceedings of ECCOMAS* (2010).
42. HINDENLANG, F. J., GASSNER, G. J., AND MUNZ, C.-D. Improving the accuracy of discontinuous Galerkin schemes at boundary layers. *submitted to International Journal on Numerical Methods for Fluid Dynamics*.
43. KIRBY, R., AND KARNIADAKIS, G. De-aliasing on non-uniform grids: algorithms and applications. *Journal of Computational Physics* 191 (2003), 249–264.
44. KLAIJ, C., VAN DER VEGT, J. J. W., AND VAN DER VEN, H. Spacetime discontinuous Galerkin method for the compressible Navier–Stokes equations. *J. Comput. Phys.* 217, 2 (2006), 589–611.
45. KLÖCKNER, A., WARBURTON, T., AND HESTHAVEN, J. S. Viscous Shock Capturing in a Time-Explicit Discontinuous Galerkin Method. *ArXiv e-prints* (Feb. 2011).
46. KOPRIVA, D. Metric identities and the discontinuous spectral element method on curvilinear meshes. *Journal of Scientific Computing* 26, 3 (2006), 301–327.
47. KOPRIVA, D. A. *Implementing Spectral Methods for Partial Differential Equations: Algorithms for Scientists and Engineers*, 1st ed. Springer Publishing Company, Incorporated, 2009.
48. KRAVCHENKO, A. G., AND MOIN, P. Numerical studies of flow over a circular cylinder at $Re_D=3900$. *Physics of Fluids* 12 (Feb. 2000), 403–417.
49. LENORMAND, E., SAGAUT, P., AND TA PHUOC, L. Large eddy simulation of subsonic and supersonic channel flow at moderate Reynolds number. *International Journal for Numerical Methods in Fluids* 32, 4 (2000), 369–406.
50. LESAINT, P., AND RAVIART, P. *On a finite element method for solving the neutron transport equation*. Academic Press, 1974.
51. LÖRCHER, F., GASSNER, G., AND MUNZ, C.-D. An explicit discontinuous Galerkin scheme with local time-stepping for general unsteady diffusion equations. *J. Comput. Phys.* 227, 11 (2008), 5649–5670.

52. LYSSENKO, D., ERTESVG, I., AND RIAN, K. Large-eddy simulation of the flow over a circular cylinder at reynolds number 3900 using the openfoam toolbox. *Flow, Turbulence and Combustion* 89, 4 (2012), 491–518.
53. MA, X., KARAMANOS, G.-S., AND KARNIADAKIS, G. E. Dynamics and low-dimensionality of a turbulent near wake. *Journal of Fluid Mechanics* 410 (5 2000), 29–65.
54. MA, X., KARAMANOS, G.-S., AND KARNIADAKIS, G. E. Dynamics and low-dimensionality of a turbulent near wake. *Journal of Fluid Mechanics* 410 (5 2000), 29–65.
55. MALM, J., SCHLATTER, P., FISCHER, P., AND HENNINGSON, D. Stabilization of the spectral element method in convection dominated flows by recovery of skew-symmetry. *Journal of Scientific Computing* 57 (2013), 254–277.
56. MASCARENHAS, B. S., HELENBROOK, B. T., AND ATKINS, H. L. Application of p-multigrid to discontinuous Galerkin formulations of the euler equations. *AIAA journal* 47, 5 (2009), 1200–1208.
57. MELLEN, C. P., FRÖHLICH, J., AND RODI, W. Large eddy simulations of the flow over periodic hills. In *IMACS World Congress* (2000), M. Deville and R. Owens, Eds.
58. MEYER, M., HICKEL, S., AND ADAMS, N. Assessment of implicit large-eddy simulation with a conservative immersed interface method for turbulent cylinder flow. *International Journal of Heat and Fluid Flow* 31, 3 (2010), 368 – 377.
59. NGUYEN, N. C., PERSSON, P.-O., AND PERAIRE, J. RANS solutions using high order discontinuous Galerkin methods. *AIAA Paper 914* (2007), 2007.
60. OL, M. V., MCAULIFFE, B. R., HANFF, E. S., SCHOLZ, U., AND KÄHLER, C. Comparison of laminar separation bubble measurements on a low Reynolds number airfoil in three facilities. *AIAA paper 5149*, 1 (2005), 2005.
61. ORSZAG, S. A. On the elimination of aliasing in finite-difference schemes by filtering high-wavenumber components. *Journal of the Atmospheric Sciences* 28, 6 (Sep 1971), 1074–1074.
62. OUVARD, H., KOOBUS, B., DERVIEUX, A., AND SALVETTI, M. V. Classical and variational multiscale LES of the flow around a circular cylinder on unstructured grids. *Computers and Fluids* 39, 7 (2010), 1083 – 1094.
63. PARNAUDEAU, P., CARLIER, J., HEITZ, D., AND LAMBALLAIS, E. Experimental and numerical studies of the flow over a circular cylinder at Reynolds number 3900. *Physics of Fluids* 20 (2008), 85–101.
64. PERAIRE, J., AND PERSSON, P. The compact discontinuous Galerkin (CDG) method for elliptic problems. *SIAM J. Sci. Comput.* 30, 4 (2008), 1806–1824.
65. REED, W., AND HILL, T. Triangular mesh methods for the neutron transport equation. Technical Report LA-UR-73-479, Los Alamos Scientific Laboratory, 1973.
66. SELIG, M. S., GUGLIELMO, J. J., BROEREN, A. P., AND GIGUERE, P. *Summary of Low-Speed Airfoil Data*, vol. 1. Soartech Publications, 1995.
67. TAYLOR, G. I. The spectrum of turbulence. *Proceedings of the Royal Society of London. Series A-Mathematical and Physical Sciences* 164, 919 (1938), 476–490.
68. TEMMERMAN, L., LESCHZINER, M. A., MELLEN, C. P., AND FRÖHLICH, J. Investigation of wall-function approximations and subgrid-scale models in large eddy simulation of separated flow in a channel with streamwise periodic constrictions. *International Journal of Heat and Fluid Flow* 24, 2 (2003), 157 – 180.
69. TORO, E. *Riemann Solvers and Numerical Methods for Fluid Dynamics*. Springer, June 1999.
70. URANGA, A., PERSSON, P.-O., DRELA, M., AND PERAIRE, J. Implicit large eddy simulation of transition to turbulence at low Reynolds numbers using a discontinuous Galerkin method. *International Journal for Numerical Methods in Engineering* 87, 1-5 (2011), 232–261.
71. WANG, Z., FIDKOWSKI, K., ABGRALL, R., BASSI, F., CARAENI, D., CARY, A., DECONINCK, H., HARTMANN, R., HILLEWAERT, K., HUYNH, H., KROLL, N., MAY, G., PERSSON, P.-O., VAN LEER, B., AND VISBAL, M. High-order cfd methods: current status and perspective. *International Journal for Numerical Methods in Fluids* 72, 8 (2013), 811–845.